

3

DERIVATIVES



Figure 3.1 The Hennessey Venom GT can go from 0 to 200 mph in 14.51 seconds. (credit: modification of work by Codex41, Flickr)

Chapter Outline

- 3.1 Defining the Derivative
- 3.2 The Derivative as a Function
- 3.3 Differentiation Rules
- 3.4 Derivatives as Rates of Change
- 3.5 Derivatives of Trigonometric Functions
- 3.6 The Chain Rule
- 3.7 Derivatives of Inverse Functions
- 3.8 Implicit Differentiation
- 3.9 Derivatives of Exponential and Logarithmic Functions



Introduction

The Hennessey Venom GT is one of the fastest cars in the world. In 2014, it reached a record-setting speed of 270.49 mph. It can go from 0 to 200 mph in 14.51 seconds. The techniques in this chapter can be used to calculate the acceleration the Venom achieves in this feat (see [Example 3.8](#).)

Calculating velocity and changes in velocity are important uses of calculus, but it is far more widespread than that. Calculus is important in all branches of mathematics, science, and engineering, and it is critical to analysis in business and health as well. In this chapter, we explore one of the main tools of calculus, the derivative, and show convenient ways to calculate derivatives. We apply these rules to a variety of functions in this chapter so that we can then explore applications of these techniques.

3.1 Defining the Derivative

Learning Objectives

- 3.1.1 Recognize the meaning of the tangent to a curve at a point.
- 3.1.2 Calculate the slope of a tangent line.
- 3.1.3 Identify the derivative as the limit of a difference quotient.
- 3.1.4 Calculate the derivative of a given function at a point.
- 3.1.5 Describe the velocity as a rate of change.
- 3.1.6 Explain the difference between average velocity and instantaneous velocity.
- 3.1.7 Estimate the derivative from a table of values.

Now that we have both a conceptual understanding of a limit and the practical ability to compute limits, we have established the foundation for our study of calculus, the branch of mathematics in which we compute derivatives and integrals. Most mathematicians and historians agree that calculus was developed independently by the Englishman Isaac Newton (1643–1727) and the German Gottfried Leibniz (1646–1716), whose images appear in [Figure 3.2](#). When we credit Newton and Leibniz with developing calculus, we are really referring to the fact that Newton and Leibniz were the first to understand the relationship between the derivative and the integral. Both mathematicians benefited from the work of predecessors, such as Barrow, Fermat, and Cavalieri. The initial relationship between the two mathematicians appears to have been amicable; however, in later years a bitter controversy erupted over whose work took precedence. Although it seems likely that Newton did, indeed, arrive at the ideas behind calculus first, we are indebted to Leibniz for the notation that we commonly use today.



Figure 3.2 Newton and Leibniz are credited with developing calculus independently.

Tangent Lines

We begin our study of calculus by revisiting the notion of secant lines and tangent lines. Recall that we used the slope of a secant line to a function at a point $(a, f(a))$ to estimate the rate of change, or the rate at which one variable changes in relation to another variable. We can obtain the slope of the secant by choosing a value of x near a and drawing a line through the points $(a, f(a))$ and $(x, f(x))$, as shown in [Figure 3.3](#). The slope of this line is given by an equation in the form of a difference quotient:

$$m_{\text{sec}} = \frac{f(x) - f(a)}{x - a}.$$

We can also calculate the slope of a secant line to a function at a value a by using this equation and replacing x with $a + h$, where h is a value close to 0. We can then calculate the slope of the line through the points $(a, f(a))$ and $(a + h, f(a + h))$. In this case, we find the secant line has a slope given by the following difference quotient with increment h :

$$m_{\text{sec}} = \frac{f(a + h) - f(a)}{a + h - a} = \frac{f(a + h) - f(a)}{h}.$$

Definition

Let f be a function defined on an interval I containing a . If $x \neq a$ is in I , then

$$Q = \frac{f(x) - f(a)}{x - a} \quad (3.1)$$

is a **difference quotient**.

Also, if $h \neq 0$ is chosen so that $a + h$ is in I , then

$$Q = \frac{f(a + h) - f(a)}{h} \quad (3.2)$$

is a difference quotient with increment h .

► MEDIA

View the development of the [derivative \(http://www.openstax.org/l/20_calcapplets\)](http://www.openstax.org/l/20_calcapplets) with this applet.

These two expressions for calculating the slope of a secant line are illustrated in [Figure 3.3](#). We will see that each of these two methods for finding the slope of a secant line is of value. Depending on the setting, we can choose one or the other. The primary consideration in our choice usually depends on ease of calculation.

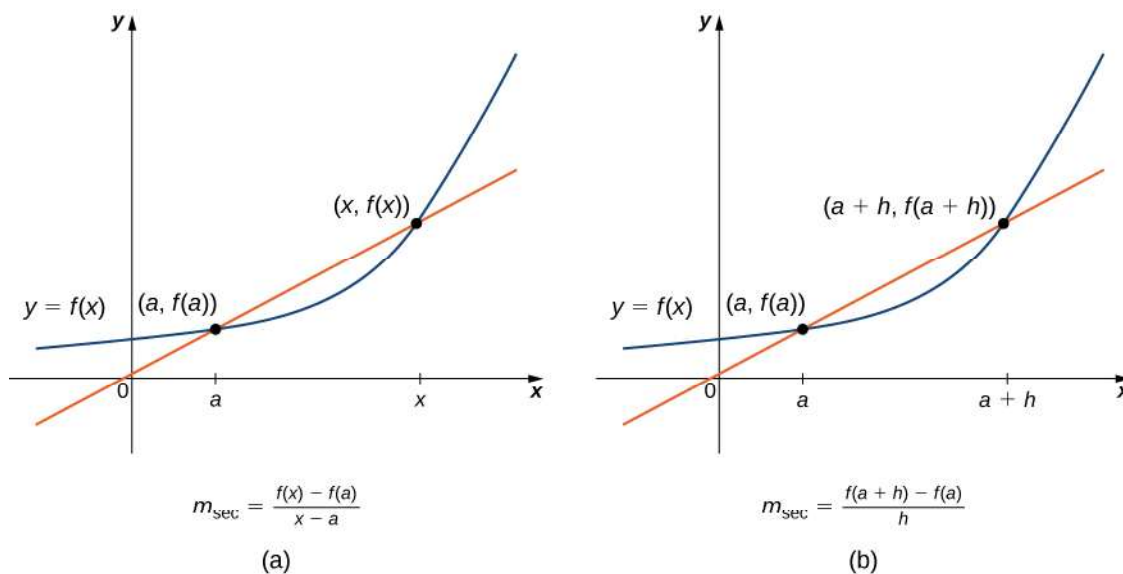


Figure 3.3 We can calculate the slope of a secant line in either of two ways.

In [Figure 3.4\(a\)](#) we see that, as the values of x approach a , the slopes of the secant lines provide better estimates of the rate of change of the function at a . Furthermore, the secant lines themselves approach the tangent line to the function at a , which represents the limit of the secant lines. Similarly, [Figure 3.4\(b\)](#) shows that as the values of h get closer to 0, the secant lines also approach the tangent line. The slope of the tangent line at a is the rate of change of the function at a , as shown in [Figure 3.4\(c\)](#).

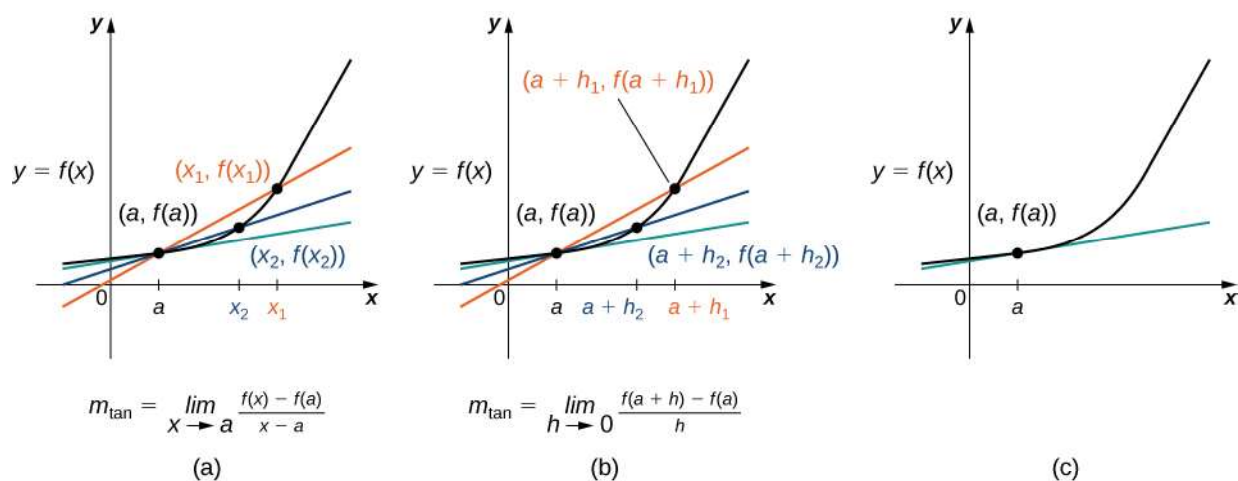


Figure 3.4 The secant lines approach the tangent line (shown in green) as the second point approaches the first.

► MEDIA

You can use this [site \(http://www.openstax.org/l/20_diffmicros\)](http://www.openstax.org/l/20_diffmicros) to explore graphs to see if they have a tangent line at a point.

In [Figure 3.5](#) we show the graph of $f(x) = \sqrt{x}$ and its tangent line at $(1, 1)$ in a series of tighter intervals about $x = 1$. As the intervals become narrower, the graph of the function and its tangent line appear to coincide, making the values on the tangent line a good approximation to the values of the function for choices of x close to 1. In fact, the graph of $f(x)$ itself appears to be locally linear in the immediate vicinity of $x = 1$.

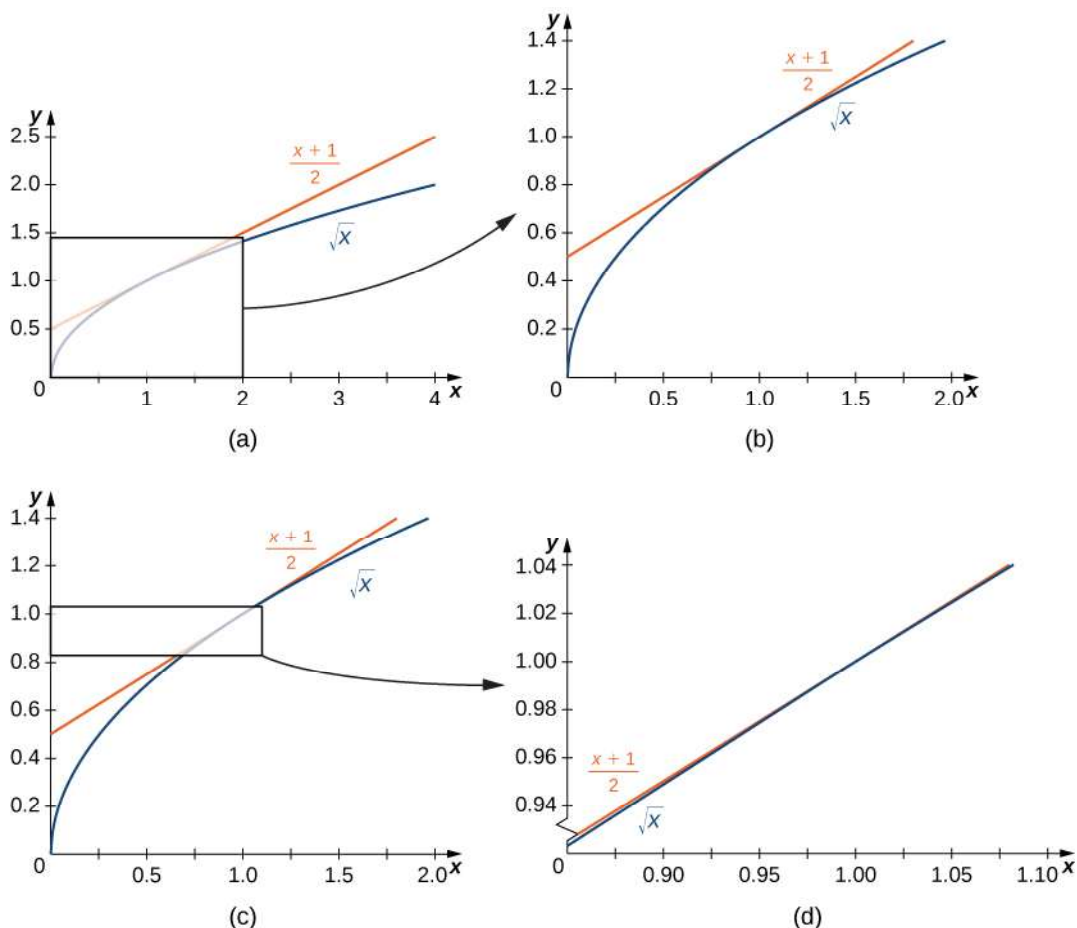


Figure 3.5 For values of x close to 1, the graph of $f(x) = \sqrt{x}$ and its tangent line appear to coincide.

Formally we may define the tangent line to the graph of a function as follows.

Definition

Let $f(x)$ be a function defined in an open interval containing a . The *tangent line* to $f(x)$ at a is the line passing through the point $(a, f(a))$ having slope

$$m_{\tan} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad (3.3)$$

provided this limit exists.

Equivalently, we may define the tangent line to $f(x)$ at a to be the line passing through the point $(a, f(a))$ having slope

$$m_{\tan} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad (3.4)$$

provided this limit exists.

Just as we have used two different expressions to define the slope of a secant line, we use two different forms to define the slope of the tangent line. In this text we use both forms of the definition. As before, the choice of definition will depend on the setting. Now that we have formally defined a tangent line to a function at a point, we can use this definition to find equations of tangent lines.

EXAMPLE 3.1

Finding a Tangent Line

Find an equation of the line tangent to the graph of $f(x) = x^2$ at $x = 3$.

✓ Solution

First find the slope of the tangent line. In this example, use [Equation 3.3](#).

$$\begin{aligned} m_{\tan} &= \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} && \text{Apply the definition.} \\ &= \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} && \text{Substitute } f(x) = x^2 \text{ and } f(3) = 9. \\ &= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3} = \lim_{x \rightarrow 3} (x+3) = 6 && \text{Factor the numerator to evaluate the limit.} \end{aligned}$$

Next, find a point on the tangent line. Since the line is tangent to the graph of $f(x)$ at $x = 3$, it passes through the point $(3, f(3))$. We have $f(3) = 9$, so the tangent line passes through the point $(3, 9)$.

Using the point-slope equation of the line with the slope $m = 6$ and the point $(3, 9)$, we obtain the line $y - 9 = 6(x - 3)$. Simplifying, we have $y = 6x - 9$. The graph of $f(x) = x^2$ and its tangent line at 3 are shown in [Figure 3.6](#).

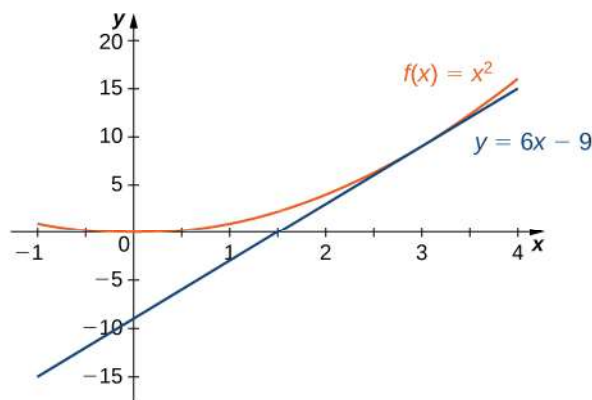


Figure 3.6 The tangent line to $f(x)$ at $x = 3$.

EXAMPLE 3.2**The Slope of a Tangent Line Revisited**

Use [Equation 3.4](#) to find the slope of the line tangent to the graph of $f(x) = x^2$ at $x = 3$.

✓ Solution

The steps are very similar to [Example 3.1](#). See [Equation 3.4](#) for the definition.

$$\begin{aligned}
 m_{\tan} &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} && \text{Apply the definition.} \\
 &= \lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h} && \text{Substitute } f(3+h) = (3+h)^2 \text{ and } f(3) = 9. \\
 &= \lim_{h \rightarrow 0} \frac{9+6h+h^2-9}{h} && \text{Expand and simplify to evaluate the limit.} \\
 &= \lim_{h \rightarrow 0} \frac{h(6+h)}{h} = \lim_{h \rightarrow 0} (6+h) = 6
 \end{aligned}$$

We obtained the same value for the slope of the tangent line by using the other definition, demonstrating that the formulas can be interchanged.

EXAMPLE 3.3**Finding the Equation of a Tangent Line**

Find an equation of the line tangent to the graph of $f(x) = 1/x$ at $x = 2$.

✓ Solution

We can use [Equation 3.3](#), but as we have seen, the results are the same if we use [Equation 3.4](#).

$$\begin{aligned}
 m_{\tan} &= \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} && \text{Apply the definition.} \\
 &= \lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2} && \text{Substitute } f(x) = \frac{1}{x} \text{ and } f(2) = \frac{1}{2}. \\
 &= \lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2} \cdot \frac{2x}{2x} && \text{Multiply numerator and denominator by } 2x \text{ to} \\
 & && \text{simplify fractions.} \\
 &= \lim_{x \rightarrow 2} \frac{(2-x)}{(x-2)(2x)} && \text{Simplify.} \\
 &= \lim_{x \rightarrow 2} \frac{-1}{2x} && \text{Simplify using } \frac{2-x}{x-2} = -1, \text{ for } x \neq 2. \\
 &= -\frac{1}{4} && \text{Evaluate the limit.}
 \end{aligned}$$

We now know that the slope of the tangent line is $-\frac{1}{4}$. To find an equation of the tangent line, we also need a point on the line. We know that $f(2) = \frac{1}{2}$. Since the tangent line passes through the point $(2, \frac{1}{2})$ we can use the point-slope equation of a line to find an equation of the tangent line. Thus the tangent line has the equation $y = -\frac{1}{4}x + 1$. The graphs of $f(x) = \frac{1}{x}$ and $y = -\frac{1}{4}x + 1$ are shown in [Figure 3.7](#).

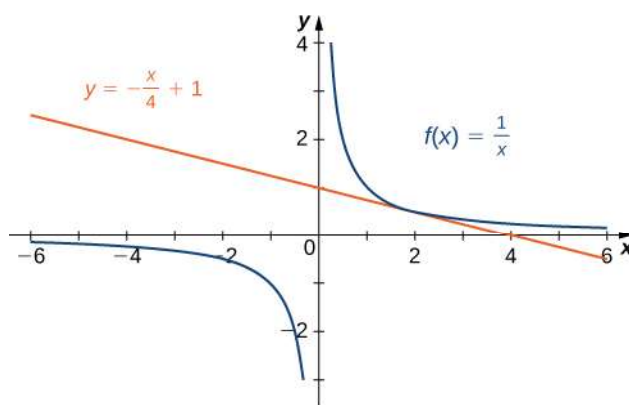


Figure 3.7 The line is tangent to $f(x)$ at $x = 2$.

- ✓ 3.1 Find the slope of the line tangent to the graph of $f(x) = \sqrt{x}$ at $x = 4$.

The Derivative of a Function at a Point

The type of limit we compute in order to find the slope of the line tangent to a function at a point occurs in many applications across many disciplines. These applications include velocity and acceleration in physics, marginal profit functions in business, and growth rates in biology. This limit occurs so frequently that we give this value a special name: the **derivative**. The process of finding a derivative is called **differentiation**.

Definition

Let $f(x)$ be a function defined in an open interval containing a . The derivative of the function $f(x)$ at a , denoted by $f'(a)$, is defined by

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad (3.5)$$

provided this limit exists.

Alternatively, we may also define the derivative of $f(x)$ at a as

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}. \quad (3.6)$$

EXAMPLE 3.4

Estimating a Derivative

For $f(x) = x^2$, use a table to estimate $f'(3)$ using [Equation 3.5](#).

✓ Solution

Create a table using values of x just below 3 and just above 3.

x	$\frac{x^2 - 9}{x - 3}$
2.9	5.9
2.99	5.99

x	$\frac{x^2-9}{x-3}$
2.999	5.999
3.001	6.001
3.01	6.01
3.1	6.1

After examining the table, we see that a good estimate is $f'(3) = 6$.

✓ 3.2 For $f(x) = x^2$, use a table to estimate $f'(3)$ using [Equation 3.6](#).

EXAMPLE 3.5

Finding a Derivative

For $f(x) = 3x^2 - 4x + 1$, find $f'(2)$ by using [Equation 3.5](#).

✓ Solution

Substitute the given function and value directly into the equation.

$$\begin{aligned}
 f'(2) &= \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} && \text{Apply the definition.} \\
 &= \lim_{x \rightarrow 2} \frac{(3x^2 - 4x + 1) - 5}{x - 2} && \text{Substitute } f(x) = 3x^2 - 4x + 1 \text{ and } f(2) = 5. \\
 &= \lim_{x \rightarrow 2} \frac{(x-2)(3x+2)}{x-2} && \text{Simplify and factor the numerator.} \\
 &= \lim_{x \rightarrow 2} (3x + 2) && \text{Cancel the common factor.} \\
 &= 8 && \text{Evaluate the limit.}
 \end{aligned}$$

EXAMPLE 3.6

Revisiting the Derivative

For $f(x) = 3x^2 - 4x + 1$, find $f'(2)$ by using [Equation 3.6](#).

✓ Solution

Using this equation, we can substitute two values of the function into the equation, and we should get the same value as in [Example 3.5](#).

$$\begin{aligned}
 f'(2) &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} && \text{Apply the definition.} \\
 &= \lim_{h \rightarrow 0} \frac{(3(2+h)^2 - 4(2+h) + 1) - 5}{h} && \text{Substitute } f(2) = 5 \text{ and } f(2+h) = 3(2+h)^2 - 4(2+h) + 1. \\
 &= \lim_{h \rightarrow 0} \frac{3h^2 + 8h}{h} && \text{Simplify the numerator.} \\
 &= \lim_{h \rightarrow 0} \frac{h(3h+8)}{h} && \text{Factor the numerator.} \\
 &= \lim_{h \rightarrow 0} (3h + 8) && \text{Cancel the common factor.} \\
 &= 8 && \text{Evaluate the limit.}
 \end{aligned}$$

The results are the same whether we use [Equation 3.5](#) or [Equation 3.6](#).

✓ 3.3 For $f(x) = x^2 + 3x + 2$, find $f'(1)$.

Velocities and Rates of Change

Now that we can evaluate a derivative, we can use it in velocity applications. Recall that if $s(t)$ is the position of an object moving along a coordinate axis, the average velocity of the object over a time interval $[a, t]$ if $t > a$ or $[t, a]$ if $t < a$ is given by the difference quotient

$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}. \quad (3.7)$$

As the values of t approach a , the values of v_{ave} approach the value we call the instantaneous velocity at a . That is, instantaneous velocity at a , denoted $v(a)$, is given by

$$v(a) = s'(a) = \lim_{t \rightarrow a} \frac{s(t) - s(a)}{t - a}. \quad (3.8)$$

To better understand the relationship between average velocity and instantaneous velocity, see [Figure 3.8](#). In this figure, the slope of the tangent line (shown in red) is the instantaneous velocity of the object at time $t = a$ whose position at time t is given by the function $s(t)$. The slope of the secant line (shown in green) is the average velocity of the object over the time interval $[a, t]$.

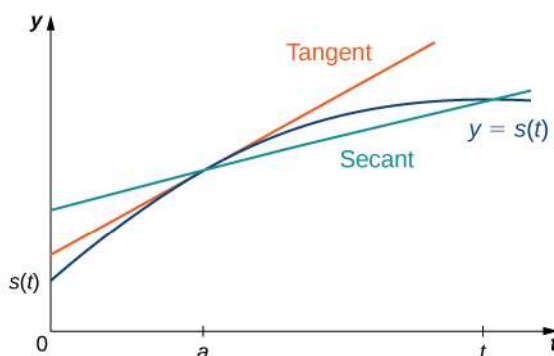


Figure 3.8 The slope of the secant line is the average velocity over the interval $[a, t]$. The slope of the tangent line is the instantaneous velocity.

We can use [Equation 3.5](#) to calculate the instantaneous velocity, or we can estimate the velocity of a moving object by using a table of values. We can then confirm the estimate by using [Equation 3.7](#).

EXAMPLE 3.7

Estimating Velocity

A lead weight on a spring is oscillating up and down. Its position at time t with respect to a fixed horizontal line is given by $s(t) = \sin t$ ([Figure 3.9](#)). Use a table of values to estimate $v(0)$. Check the estimate by using [Equation 3.5](#).

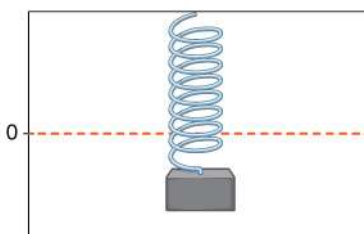


Figure 3.9 A lead weight suspended from a spring in vertical oscillatory motion.

✓ Solution

We can estimate the instantaneous velocity at $t = 0$ by computing a table of average velocities using values of t

approaching 0, as shown in [Table 3.1](#).

t	$\frac{\sin t - \sin 0}{t - 0} = \frac{\sin t}{t}$
-0.1	0.998334166
-0.01	0.9999833333
-0.001	0.999999833
0.001	0.999999833
0.01	0.9999833333
0.1	0.998334166

Table 3.1 Average velocities using values of t approaching 0

From the table we see that the average velocity over the time interval $[-0.1, 0]$ is 0.998334166, the average velocity over the time interval $[-0.01, 0]$ is 0.9999833333, and so forth. Using this table of values, it appears that a good estimate is $v(0) = 1$.

By using [Equation 3.5](#), we can see that

$$v(0) = s'(0) = \lim_{t \rightarrow 0} \frac{\sin t - \sin 0}{t - 0} = \lim_{t \rightarrow 0} \frac{\sin t}{t} = 1.$$

Thus, in fact, $v(0) = 1$.

-
- ✓ 3.4 A rock is dropped from a height of 64 feet. Its height above ground at time t seconds later is given by $s(t) = -16t^2 + 64$, $0 \leq t \leq 2$. Find its instantaneous velocity 1 second after it is dropped, using [Equation 3.5](#).

As we have seen throughout this section, the slope of a tangent line to a function and instantaneous velocity are related concepts. Each is calculated by computing a derivative and each measures the instantaneous rate of change of a function, or the rate of change of a function at any point along the function.

Definition

The **instantaneous rate of change** of a function $f(x)$ at a value a is its derivative $f'(a)$.

EXAMPLE 3.8**Chapter Opener: Estimating Rate of Change of Velocity****Figure 3.10** (credit: modification of work by Codex41, Flickr)

Reaching a top speed of 270.49 mph, the Hennessey Venom GT is one of the fastest cars in the world. In tests it went from 0 to 60 mph in 3.05 seconds, from 0 to 100 mph in 5.88 seconds, from 0 to 200 mph in 14.51 seconds, and from 0 to 229.9 mph in 19.96 seconds. Use this data to draw a conclusion about the rate of change of velocity (that is, its acceleration) as it approaches 229.9 mph. Does the rate at which the car is accelerating appear to be increasing, decreasing, or constant?

✓ Solution

First observe that 60 mph = 88 ft/s, 100 mph $\approx$ 146.67 ft/s, 200 mph $\approx$ 293.33 ft/s, and 229.9 mph $\approx$ 337.19 ft/s. We can summarize the information in a table.

t	$v(t)$
0	0
3.05	88
5.88	146.67
14.51	293.33
19.96	337.19

Table 3.2 $v(t)$ at different values of t

Now compute the average acceleration of the car in feet per second per second on intervals of the form $[t, 19.96]$ as t approaches 19.96, as shown in the following table.

t	$\frac{v(t)-v(19.96)}{t-19.96} = \frac{v(t)-337.19}{t-19.96}$
0.0	16.89
3.05	14.74
5.88	13.53
14.51	8.05

Table 3.3 Average acceleration

The rate at which the car is accelerating is decreasing as its velocity approaches 229.9 mph (337.19 ft/s).

EXAMPLE 3.9**Rate of Change of Temperature**

A homeowner sets the thermostat so that the temperature in the house begins to drop from 70°F at 9 p.m., reaches a low of 60° during the night, and rises back to 70° by 7 a.m. the next morning. Suppose that the temperature in the house is given by $T(t) = 0.4t^2 - 4t + 70$ for $0 \leq t \leq 10$, where t is the number of hours past 9 p.m. Find the instantaneous rate of change of the temperature at midnight.

✓ Solution

Since midnight is 3 hours past 9 p.m., we want to compute $T'(3)$. Refer to [Equation 3.5](#).

$$\begin{aligned}
 T'(3) &= \lim_{t \rightarrow 3} \frac{T(t) - T(3)}{t - 3} && \text{Apply the definition.} \\
 &= \lim_{t \rightarrow 3} \frac{0.4t^2 - 4t + 70 - 61.6}{t - 3} && \text{Substitute } T(t) = 0.4t^2 - 4t + 70 \text{ and } T(3) = 61.6. \\
 &= \lim_{t \rightarrow 3} \frac{0.4t^2 - 4t + 8.4}{t - 3} && \text{Simplify.} \\
 &= \lim_{t \rightarrow 3} \frac{0.4(t-3)(t-7)}{t-3} && = \lim_{t \rightarrow 3} \frac{0.4(t-3)(t-7)}{t-3} \\
 &= \lim_{t \rightarrow 3} 0.4(t-7) && \text{Cancel.} \\
 &= -1.6 && \text{Evaluate the limit.}
 \end{aligned}$$

The instantaneous rate of change of the temperature at midnight is -1.6°F per hour.

EXAMPLE 3.10**Rate of Change of Profit**

A toy company can sell x electronic gaming systems at a price of $p = -0.01x + 400$ dollars per gaming system. The cost of manufacturing x systems is given by $C(x) = 100x + 10,000$ dollars. Find the rate of change of profit when 10,000 games are produced. Should the toy company increase or decrease production?

✓ Solution

The profit $P(x)$ earned by producing x gaming systems is $R(x) - C(x)$, where $R(x)$ is the revenue obtained from the sale of x games. Since the company can sell x games at $p = -0.01x + 400$ per game,

$$R(x) = xp = x(-0.01x + 400) = -0.01x^2 + 400x.$$

Consequently,

$$P(x) = -0.01x^2 + 300x - 10,000.$$

Therefore, evaluating the rate of change of profit gives

$$\begin{aligned}
 P'(10000) &= \lim_{x \rightarrow 10000} \frac{P(x) - P(10000)}{x - 10000} \\
 &= \lim_{x \rightarrow 10000} \frac{-0.01x^2 + 300x - 10000 - 1990000}{x - 10000} \\
 &= \lim_{x \rightarrow 10000} \frac{-0.01x^2 + 300x - 2000000}{x - 10000} \\
 &= 100.
 \end{aligned}$$

Since the rate of change of profit $P'(10,000) > 0$ and $P(10,000) > 0$, the company should increase production.

- ✓ 3.5 A coffee shop determines that the daily profit on scones obtained by charging s dollars per scone is $P(s) = -20s^2 + 150s - 10$. The coffee shop currently charges \$3.25 per scone. Find $P'(3.25)$, the rate of change of profit when the price is \$3.25 and decide whether or not the coffee shop should consider raising or lowering its prices on scones.



SECTION 3.1 EXERCISES

For the following exercises, use [Equation 3.1](#) to find the slope of the secant line between the values x_1 and x_2 for each function $y = f(x)$.

1. $f(x) = 4x + 7$; $x_1 = 2, x_2 = 5$
2. $f(x) = 8x - 3$; $x_1 = -1, x_2 = 3$
3. $f(x) = x^2 + 2x + 1$; $x_1 = 3, x_2 = 3.5$
4. $f(x) = -x^2 + x + 2$; $x_1 = 0.5, x_2 = 1.5$
5. $f(x) = \frac{4}{3x-1}$; $x_1 = 1, x_2 = 3$
6. $f(x) = \frac{x-7}{2x+1}$; $x_1 = 0, x_2 = 2$
7. $f(x) = \sqrt{x}$; $x_1 = 1, x_2 = 16$
8. $f(x) = \sqrt{x-9}$; $x_1 = 10, x_2 = 13$
9. $f(x) = x^{1/3} + 1$; $x_1 = 0, x_2 = 8$
10. $f(x) = 6x^{2/3} + 2x^{1/3}$; $x_1 = 1, x_2 = 27$

For the following functions,

- a. use [Equation 3.4](#) to find the slope of the tangent line $m_{\tan} = f'(a)$, and
- b. find an equation of the tangent line to f at $x = a$.

11. $f(x) = 3 - 4x, a = 2$
12. $f(x) = \frac{x}{5} + 6, a = -1$
13. $f(x) = x^2 + x, a = 1$
14. $f(x) = 1 - x - x^2, a = 0$
15. $f(x) = \frac{7}{x}, a = 3$
16. $f(x) = \sqrt{x+8}, a = 1$
17. $f(x) = 2 - 3x^2, a = -2$
18. $f(x) = \frac{-3}{x-1}, a = 4$
19. $f(x) = \frac{2}{x+3}, a = -4$
20. $f(x) = \frac{3}{x^2}, a = 3$

For the following functions $y = f(x)$, find $f'(a)$ using [Equation 3.5](#).

21. $f(x) = 5x + 4, a = -1$
22. $f(x) = -7x + 1, a = 3$
23. $f(x) = x^2 + 9x, a = 2$
24. $f(x) = 3x^2 - x + 2, a = 1$
25. $f(x) = \sqrt{x}, a = 4$
26. $f(x) = \sqrt{x-2}, a = 6$
27. $f(x) = \frac{1}{x}, a = 2$
28. $f(x) = \frac{1}{x-3}, a = -1$
29. $f(x) = \frac{1}{x^3}, a = 1$
30. $f(x) = \frac{1}{\sqrt{x}}, a = 4$

For the following exercises, given the function $y = f(x)$,

- a. find the slope of the secant line PQ for each point $Q(x, f(x))$ with x value given in the table.
- b. Use the answers from a. to estimate the value of the slope of the tangent line at P .
- c. Use the answer from b. to find an equation of the tangent line to f at point P .

31. [T] $f(x) = x^2 + 3x + 4$, $P(1, 8)$ (Round to 6 decimal places.)

x	Slope m_{PQ}	x	Slope m_{PQ}
1.1	(i)	0.9	(vii)
1.01	(ii)	0.99	(viii)
1.001	(iii)	0.999	(ix)
1.0001	(iv)	0.9999	(x)
1.00001	(v)	0.99999	(xi)
1.000001	(vi)	0.999999	(xii)

32. [T] $f(x) = \frac{x+1}{x^2-1}$, $P(0, -1)$

x	Slope m_{PQ}	x	Slope m_{PQ}
0.1	(i)	-0.1	(vii)
0.01	(ii)	-0.01	(viii)
0.001	(iii)	-0.001	(ix)
0.0001	(iv)	-0.0001	(x)
0.00001	(v)	-0.00001	(xi)
0.000001	(vi)	-0.000001	(xii)

33. [T]
 $f(x) = 10e^{0.5x}$, $P(0, 10)$
 (Round to 4 decimal places.)

x	Slope m_{PQ}
-0.1	(i)
-0.01	(ii)
-0.001	(iii)
-0.0001	(iv)
-0.00001	(v)
-0.000001	(vi)

34. [T] $f(x) = \tan(x)$, $P(\pi, 0)$

x	Slope m_{PQ}
3.1	(i)
3.14	(ii)
3.141	(iii)
3.1415	(iv)
3.14159	(v)
3.141592	(vi)

[T] For the following position functions $y = s(t)$, an object is moving along a straight line, where t is in seconds and s is in meters. Find

- the simplified expression for the average velocity from $t = 2$ to $t = 2 + h$;
- the average velocity between $t = 2$ and $t = 2 + h$, where (i) $h = 0.1$, (ii) $h = 0.01$, (iii) $h = 0.001$, and (iv) $h = 0.0001$; and
- use the answer from a. to estimate the instantaneous velocity at $t = 2$ second.

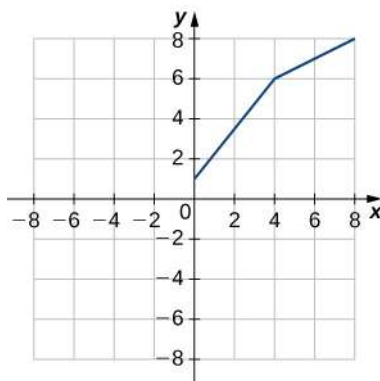
35. $s(t) = \frac{1}{3}t + 5$

36. $s(t) = t^2 - 2t$

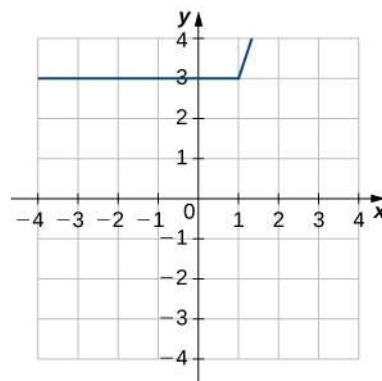
37. $s(t) = 2t^3 + 3$

38. $s(t) = \frac{16}{t^2} - \frac{4}{t}$

39. Use the following graph to evaluate a. $f'(1)$ and b. $f'(6)$.



40. Use the following graph to evaluate a. $f'(-3)$ and b. $f'(1.5)$.



For the following exercises, use the limit definition of derivative to show that the derivative does not exist at $x = a$ for each of the given functions.

41. $f(x) = x^{1/3}, x = 0$

42. $f(x) = x^{2/3}, x = 0$

43. $f(x) = \begin{cases} 1, & x < 1 \\ x, & x \geq 1 \end{cases}, x = 1$

44. $f(x) = \frac{|x|}{x}, x = 0$

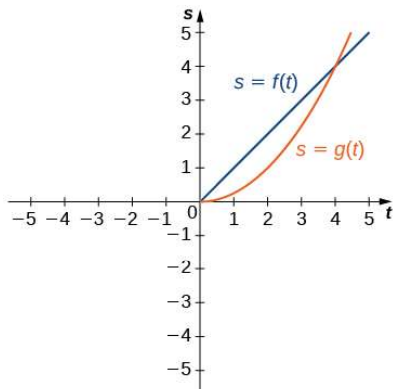
45. [T] The position in feet of a race car along a straight track after t seconds is modeled by the function $s(t) = 8t^2 - \frac{1}{16}t^3$.

- Find the average velocity of the vehicle over the following time intervals to four decimal places:
 - $[4, 4.1]$
 - $[4, 4.01]$
 - $[4, 4.001]$
 - $[4, 4.0001]$
- Use a. to draw a conclusion about the instantaneous velocity of the vehicle at $t = 4$ seconds.

46. [T] The distance in feet that a ball rolls down an incline is modeled by the function $s(t) = 14t^2$, where t is seconds after the ball begins rolling.

- Find the average velocity of the ball over the following time intervals:
 - $[5, 5.1]$
 - $[5, 5.01]$
 - $[5, 5.001]$
 - $[5, 5.0001]$
- Use the answers from a. to draw a conclusion about the instantaneous velocity of the ball at $t = 5$ seconds.

47. Two vehicles start out traveling side by side along a straight road. Their position functions, shown in the following graph, are given by $s = f(t)$ and $s = g(t)$, where s is measured in feet and t is measured in seconds.



- Which vehicle has traveled farther at $t = 2$ seconds?
 - What is the approximate velocity of each vehicle at $t = 3$ seconds?
 - Which vehicle is traveling faster at $t = 4$ seconds?
 - What is true about the positions of the vehicles at $t = 4$ seconds?
48. [T] The total cost $C(x)$, in hundreds of dollars, to produce x thousand jars of mayonnaise is given by $C(x) = 0.000003x^3 + 4x + 300$.
- Calculate the average cost per jar over the following intervals:
 - $[100, 100.1]$
 - $[100, 100.01]$
 - $[100, 100.001]$
 - $[100, 100.0001]$
 - Use the answers from a. to estimate the average cost to produce 100,000 jars of mayonnaise.
49. [T] For the function $f(x) = x^3 - 2x^2 - 11x + 12$, do the following.
- Use a graphing calculator to graph f in an appropriate viewing window.
 - Use the ZOOM feature on the calculator to approximate the two values of $x = a$ for which $m_{\tan} = f'(a) = 0$.
50. [T] For the function $f(x) = \frac{x}{1+x^2}$, do the following.
- Use a graphing calculator to graph f in an appropriate viewing window.
 - Use the ZOOM feature on the calculator to approximate the values of $x = a$ for which $m_{\tan} = f'(a) = 0$.
51. Suppose that $N(x)$ computes the number of gallons of gas used by a vehicle traveling x miles. Suppose the vehicle gets 30 mpg.
- Find a mathematical expression for $N(x)$.
 - What is $N(100)$? Explain the physical meaning.
 - What is $N'(100)$? Explain the physical meaning.
52. [T] For the function $f(x) = x^4 - 5x^2 + 4$, do the following.
- Use a graphing calculator to graph f in an appropriate viewing window.
 - Use the nDeriv function, which numerically finds the derivative, on a graphing calculator to estimate $f'(-2)$, $f'(-0.5)$, $f'(1.7)$, and $f'(2.718)$.
53. [T] For the function $f(x) = \frac{x^2}{x^2+1}$, do the following.
- Use a graphing calculator to graph f in an appropriate viewing window.
 - Use the nDeriv function on a graphing calculator to find $f'(-4)$, $f'(-2)$, $f'(2)$, and $f'(4)$.

3.2 The Derivative as a Function

Learning Objectives

- 3.2.1 Define the derivative function of a given function.
- 3.2.2 Graph a derivative function from the graph of a given function.
- 3.2.3 State the connection between derivatives and continuity.
- 3.2.4 Describe three conditions for when a function does not have a derivative.
- 3.2.5 Explain the meaning of a higher-order derivative.

As we have seen, the derivative of a function at a given point gives us the rate of change or slope of the tangent line to the function at that point. If we differentiate a position function at a given time, we obtain the velocity at that time. It seems reasonable to conclude that knowing the derivative of the function at every point would produce valuable information about the behavior of the function. However, the process of finding the derivative at even a handful of values using the techniques of the preceding section would quickly become quite tedious. In this section we define the derivative function and learn a process for finding it.

Derivative Functions

The derivative function gives the derivative of a function at each point in the domain of the original function for which the derivative is defined. We can formally define a derivative function as follows.

Definition

Let f be a function. The **derivative function**, denoted by f' , is the function whose domain consists of those values of x such that the following limit exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}. \quad (3.9)$$

A function $f(x)$ is said to be **differentiable at a** if $f'(a)$ exists. More generally, a function is said to be **differentiable on S** if it is differentiable at every point in an open set S , and a **differentiable function** is one in which $f'(x)$ exists on its domain.

In the next few examples we use Equation 3.9 to find the derivative of a function.

EXAMPLE 3.11

Finding the Derivative of a Square-Root Function

Find the derivative of $f(x) = \sqrt{x}$.

✓ Solution

Start directly with the definition of the derivative function. Use Equation 3.1.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{1}{(\sqrt{x+h} + \sqrt{x})} \\ &= \frac{1}{2\sqrt{x}} \end{aligned}$$

Substitute $f(x+h) = \sqrt{x+h}$ and $f(x) = \sqrt{x}$ into $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.

Multiply numerator and denominator by $\sqrt{x+h} + \sqrt{x}$ without distributing in the denominator.

Multiply the numerators and simplify.

Cancel the h .

Evaluate the limit.

EXAMPLE 3.12**Finding the Derivative of a Quadratic Function**

Find the derivative of the function $f(x) = x^2 - 2x$.

✓ Solution

Follow the same procedure here, but without having to multiply by the conjugate.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{((x+h)^2 - 2(x+h)) - (x^2 - 2x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h} \\ &= \lim_{h \rightarrow 0} \frac{2xh - 2h + h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2x - 2 + h)}{h} \\ &= \lim_{h \rightarrow 0} (2x - 2 + h) \\ &= 2x - 2 \end{aligned}$$

Substitute $f(x+h) = (x+h)^2 - 2(x+h)$ and

$f(x) = x^2 - 2x$ into

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Expand $(x+h)^2 - 2(x+h)$.

Simplify.

Factor out h from the numerator.

Cancel the common factor of h .

Evaluate the limit.

✓ 3.6 Find the derivative of $f(x) = x^2$.

We use a variety of different notations to express the derivative of a function. In [Example 3.12](#) we showed that if $f(x) = x^2 - 2x$, then $f'(x) = 2x - 2$. If we had expressed this function in the form $y = x^2 - 2x$, we could have expressed the derivative as $y' = 2x - 2$ or $\frac{dy}{dx} = 2x - 2$. We could have conveyed the same information by writing $\frac{d}{dx}(x^2 - 2x) = 2x - 2$. Thus, for the function $y = f(x)$, each of the following notations represents the derivative of $f(x)$:

$$f'(x), \frac{dy}{dx}, y', \frac{d}{dx}(f(x)).$$

In place of $f'(a)$ we may also use $\left. \frac{dy}{dx} \right|_{x=a}$. Use of the $\frac{dy}{dx}$ notation (called Leibniz notation) is quite common in engineering and physics. To understand this notation better, recall that the derivative of a function at a point is the limit of the slopes of secant lines as the secant lines approach the tangent line. The slopes of these secant lines are often expressed in the form $\frac{\Delta y}{\Delta x}$ where Δy is the difference in the y values corresponding to the difference in the x values, which are expressed as Δx ([Figure 3.11](#)). Thus the derivative, which can be thought of as the instantaneous rate of change of y with respect to x , is expressed as

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}.$$

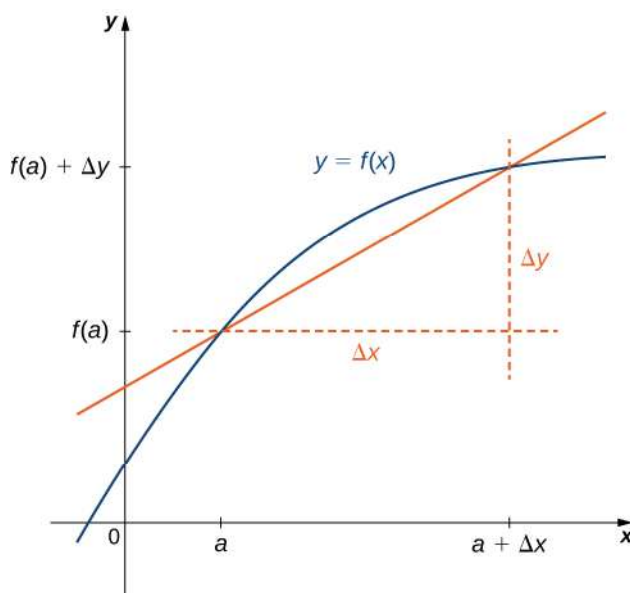


Figure 3.11 The derivative is expressed as $\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$.

Graphing a Derivative

We have already discussed how to graph a function, so given the equation of a function or the equation of a derivative function, we could graph it. Given both, we would expect to see a correspondence between the graphs of these two functions, since $f'(x)$ gives the rate of change of a function $f(x)$ (or slope of the tangent line to $f(x)$).

In [Example 3.11](#) we found that for $f(x) = \sqrt{x}$, $f'(x) = \frac{1}{2\sqrt{x}}$. If we graph these functions on the same axes, as in

[Figure 3.12](#), we can use the graphs to understand the relationship between these two functions. First, we notice that $f(x)$ is increasing over its entire domain, which means that the slopes of its tangent lines at all points are positive. Consequently, we expect $f'(x) > 0$ for all values of x in its domain. Furthermore, as x increases, the slopes of the tangent lines to $f(x)$ are decreasing and we expect to see a corresponding decrease in $f'(x)$. We also observe that $f'(0)$ is undefined and that $\lim_{x \rightarrow 0^+} f'(x) = +\infty$, corresponding to a vertical tangent to $f(x)$ at 0.

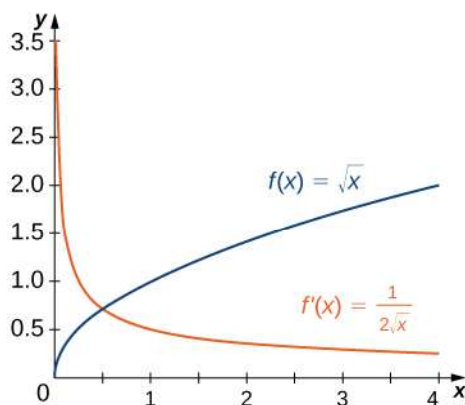


Figure 3.12 The derivative $f'(x)$ is positive everywhere because the function $f(x)$ is increasing.

In [Example 3.12](#) we found that for $f(x) = x^2 - 2x$, $f'(x) = 2x - 2$. The graphs of these functions are shown in [Figure 3.13](#). Observe that $f(x)$ is decreasing for $x < 1$. For these same values of x , $f'(x) < 0$. For values of $x > 1$, $f(x)$ is increasing and $f'(x) > 0$. Also, $f(x)$ has a horizontal tangent at $x = 1$ and $f'(1) = 0$.

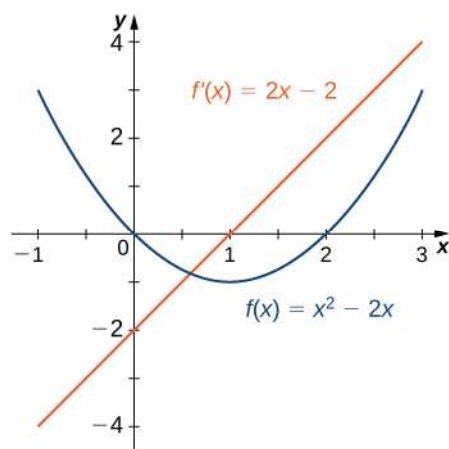
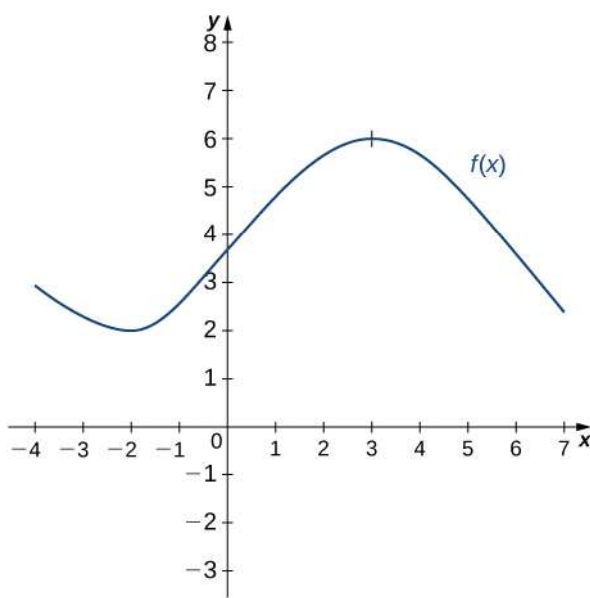


Figure 3.13 The derivative $f'(x) < 0$ where the function $f(x)$ is decreasing and $f'(x) > 0$ where $f(x)$ is increasing. The derivative is zero where the function has a horizontal tangent.

EXAMPLE 3.13

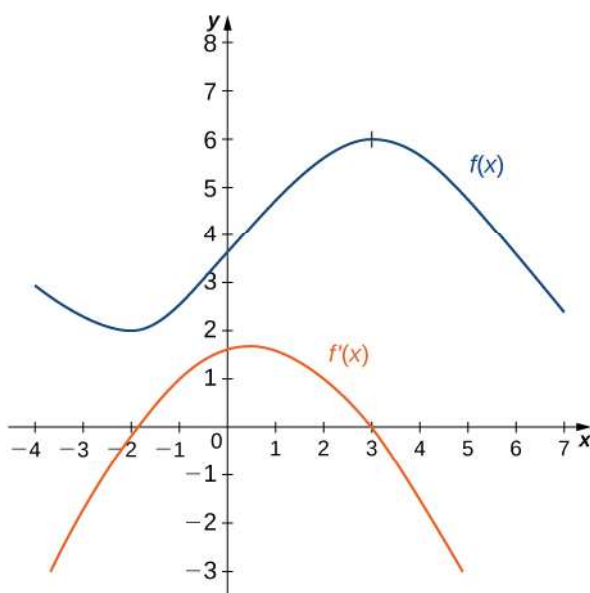
Sketching a Derivative Using a Function

Use the following graph of $f(x)$ to sketch a graph of $f'(x)$.



Solution

The solution is shown in the following graph. Observe that $f(x)$ is increasing and $f'(x) > 0$ on $(-2, 3)$. Also, $f(x)$ is decreasing and $f'(x) < 0$ on $(-\infty, -2)$ and on $(3, +\infty)$. Also note that $f(x)$ has horizontal tangents at -2 and 3 , and $f'(-2) = 0$ and $f'(3) = 0$.



- ✓ 3.7 Sketch the graph of $f(x) = x^2 - 4$. On what interval is the graph of $f'(x)$ above the x -axis?

Derivatives and Continuity

Now that we can graph a derivative, let's examine the behavior of the graphs. First, we consider the relationship between differentiability and continuity. We will see that if a function is differentiable at a point, it must be continuous there; however, a function that is continuous at a point need not be differentiable at that point. In fact, a function may be continuous at a point and fail to be differentiable at the point for one of several reasons.

Theorem 3.1

Differentiability Implies Continuity

Let $f(x)$ be a function and a be in its domain. If $f(x)$ is differentiable at a , then f is continuous at a .

Proof

If $f(x)$ is differentiable at a , then $f'(a)$ exists and

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

We want to show that $f(x)$ is continuous at a by showing that $\lim_{x \rightarrow a} f(x) = f(a)$. Thus,

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (f(x) - f(a) + f(a)) \\ &= \lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} \cdot (x - a) + f(a) \right) && \text{Multiply and divide } f(x) - f(a) \text{ by } x - a. \\ &= \left(\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right) \cdot \left(\lim_{x \rightarrow a} (x - a) \right) + \lim_{x \rightarrow a} f(a) \\ &= f'(a) \cdot 0 + f(a) \\ &= f(a). \end{aligned}$$

Therefore, since $f(a)$ is defined and $\lim_{x \rightarrow a} f(x) = f(a)$, we conclude that f is continuous at a .

□

We have just proven that differentiability implies continuity, but now we consider whether continuity implies differentiability. To determine an answer to this question, we examine the function $f(x) = |x|$. This function is

continuous everywhere; however, $f'(0)$ is undefined. This observation leads us to believe that continuity does not imply differentiability. Let's explore further. For $f(x) = |x|$,

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{|x| - |0|}{x - 0} = \lim_{x \rightarrow 0} \frac{|x|}{x}.$$

This limit does not exist because

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1 \text{ and } \lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1.$$

See [Figure 3.14](#).

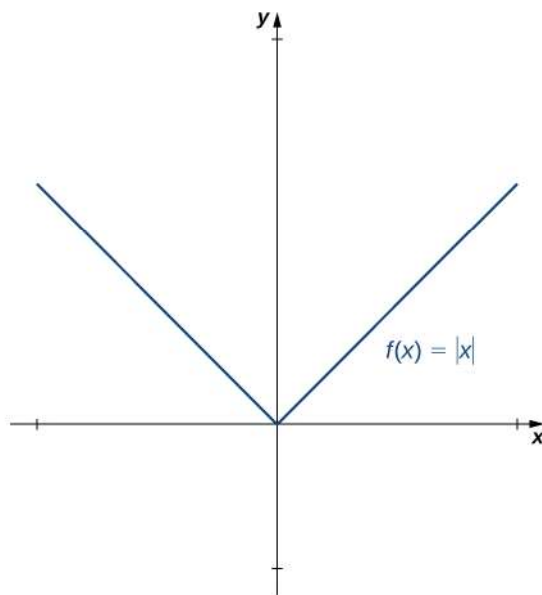


Figure 3.14 The function $f(x) = |x|$ is continuous at 0 but is not differentiable at 0.

Let's consider some additional situations in which a continuous function fails to be differentiable. Consider the function $f(x) = \sqrt[3]{x}$:

$$f'(0) = \lim_{x \rightarrow 0} \frac{\sqrt[3]{x} - 0}{x - 0} = \lim_{x \rightarrow 0} \frac{1}{\sqrt[3]{x^2}} = +\infty.$$

Thus $f'(0)$ does not exist. A quick look at the graph of $f(x) = \sqrt[3]{x}$ clarifies the situation. The function has a vertical tangent line at 0 ([Figure 3.15](#)).

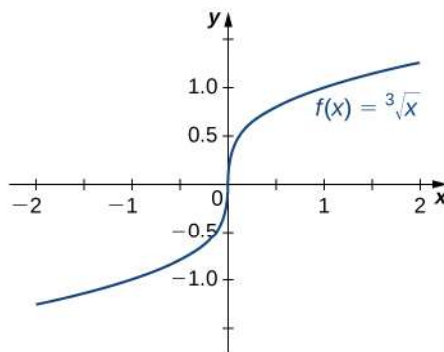


Figure 3.15 The function $f(x) = \sqrt[3]{x}$ has a vertical tangent at $x = 0$. It is continuous at 0 but is not differentiable at 0.

The function $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ also has a derivative that exhibits interesting behavior at 0. We see that

$$f'(0) = \lim_{x \rightarrow 0} \frac{x \sin(1/x) - 0}{x - 0} = \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right).$$

This limit does not exist, essentially because the slopes of the secant lines continuously change direction as they approach zero (Figure 3.16).

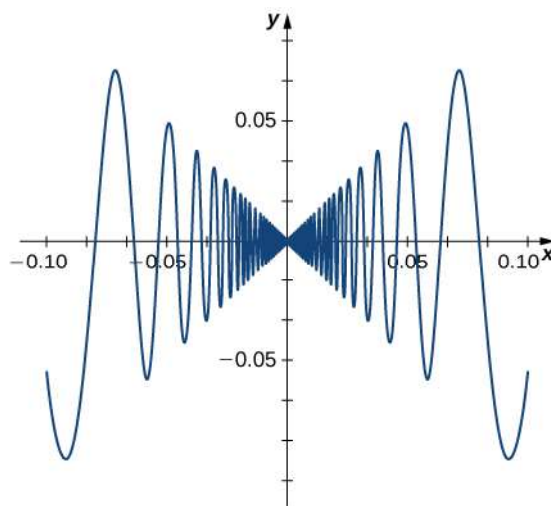


Figure 3.16 The function $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ is not differentiable at 0.

In summary:

1. We observe that if a function is not continuous, it cannot be differentiable, since every differentiable function must be continuous. However, if a function is continuous, it may still fail to be differentiable.
2. We saw that $f(x) = |x|$ failed to be differentiable at 0 because the limit of the slopes of the tangent lines on the left and right were not the same. Visually, this resulted in a sharp corner on the graph of the function at 0. From this we conclude that in order to be differentiable at a point, a function must be “smooth” at that point.
3. As we saw in the example of $f(x) = \sqrt[3]{x}$, a function fails to be differentiable at a point where there is a vertical tangent line.
4. As we saw with $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ a function may fail to be differentiable at a point in more complicated ways as well.

EXAMPLE 3.14

A Piecewise Function that is Continuous and Differentiable

A toy company wants to design a track for a toy car that starts out along a parabolic curve and then converts to a straight line (Figure 3.17). The function that describes the track is to have the form $f(x) = \begin{cases} \frac{1}{10}x^2 + bx + c & \text{if } x < -10 \\ -\frac{1}{4}x + \frac{5}{2} & \text{if } x \geq -10 \end{cases}$ where x and $f(x)$ are in inches. For the car to move smoothly along the track, the function $f(x)$ must be both continuous and differentiable at -10 . Find values of b and c that make $f(x)$ both continuous and differentiable.

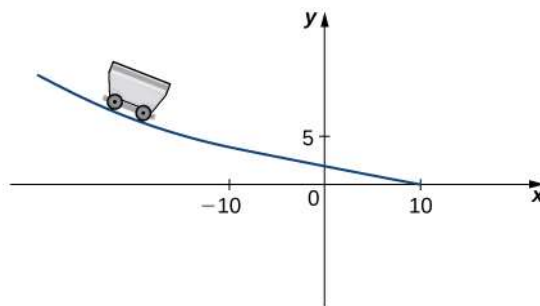


Figure 3.17 For the car to move smoothly along the track, the function must be both continuous and differentiable.

✓ **Solution**

For the function to be continuous at $x = -10$, $\lim_{x \rightarrow -10^-} f(x) = f(-10)$. Thus, since

$$\lim_{x \rightarrow -10^-} f(x) = \frac{1}{10}(-10)^2 - 10b + c = 10 - 10b + c$$

and $f(-10) = 5$, we must have $10 - 10b + c = 5$. Equivalently, we have $c = 10b - 5$.

For the function to be differentiable at -10 ,

$$f'(-10) = \lim_{x \rightarrow -10} \frac{f(x) - f(-10)}{x + 10}$$

must exist. Since $f(x)$ is defined using different rules on the right and the left, we must evaluate this limit from the right and the left and then set them equal to each other:

$$\begin{aligned} \lim_{x \rightarrow -10^-} \frac{f(x) - f(-10)}{x + 10} &= \lim_{x \rightarrow -10^-} \frac{\frac{1}{10}x^2 + bx + c - 5}{x + 10} \\ &= \lim_{x \rightarrow -10^-} \frac{\frac{1}{10}x^2 + bx + (10b - 5) - 5}{x + 10} && \text{Substitute } c = 10b - 5. \\ &= \lim_{x \rightarrow -10^-} \frac{x^2 - 100 + 10bx + 100b}{10(x + 10)} && \text{Multiply numerator and denominator by 10.} \\ &= \lim_{x \rightarrow -10^-} \frac{(x + 10)(x - 10 + 10b)}{10(x + 10)} && \text{Factor by grouping.} \\ &= b - 2. \end{aligned}$$

We also have

$$\begin{aligned} \lim_{x \rightarrow -10^+} \frac{f(x) - f(-10)}{x + 10} &= \lim_{x \rightarrow -10^+} \frac{-\frac{1}{4}x + \frac{5}{2} - 5}{x + 10} \\ &= \lim_{x \rightarrow -10^+} \frac{-(x + 10)}{4(x + 10)} \\ &= -\frac{1}{4}. \end{aligned}$$

This gives us $b - 2 = -\frac{1}{4}$. Thus $b = \frac{7}{4}$ and $c = 10\left(\frac{7}{4}\right) - 5 = \frac{25}{2}$.

✓ 3.8 Find values of a and b that make $f(x) = \begin{cases} ax + b & \text{if } x < 3 \\ x^2 & \text{if } x \geq 3 \end{cases}$ both continuous and differentiable at 3.

Higher-Order Derivatives

The derivative of a function is itself a function, so we can find the derivative of a derivative. For example, the derivative of a position function is the rate of change of position, or velocity. The derivative of velocity is the rate of change of velocity, which is acceleration. The new function obtained by differentiating the derivative is called the second derivative. Furthermore, we can continue to take derivatives to obtain the third derivative, fourth derivative, and so on. Collectively, these are referred to as **higher-order derivatives**. The notation for the higher-order derivatives of $y = f(x)$ can be

expressed in any of the following forms:

$$f''(x), f'''(x), f^{(4)}(x), \dots, f^{(n)}(x)$$

$$y''(x), y'''(x), y^{(4)}(x), \dots, y^{(n)}(x)$$

$$\frac{d^2 y}{dx^2}, \frac{d^3 y}{dx^3}, \frac{d^4 y}{dx^4}, \dots, \frac{d^n y}{dx^n}.$$

It is interesting to note that the notation for $\frac{d^2 y}{dx^2}$ may be viewed as an attempt to express $\frac{d}{dx} \left(\frac{dy}{dx} \right)$ more compactly.

Analogously, $\frac{d}{dx} \left(\frac{d}{dx} \left(\frac{dy}{dx} \right) \right) = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = \frac{d^3 y}{dx^3}$.

EXAMPLE 3.15

Finding a Second Derivative

For $f(x) = 2x^2 - 3x + 1$, find $f''(x)$.

Solution

First find $f'(x)$.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(2(x+h)^2 - 3(x+h) + 1) - (2x^2 - 3x + 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 3h}{h} \\ &= \lim_{h \rightarrow 0} (4x + 2h - 3) \\ &= 4x - 3 \end{aligned}$$

Substitute $f(x) = 2x^2 - 3x + 1$

and

$$f(x+h) = 2(x+h)^2 - 3(x+h) + 1$$

$$\text{into } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Simplify the numerator.

Factor out the h in the numerator and cancel with the h in the denominator.

Take the limit.

Next, find $f''(x)$ by taking the derivative of $f'(x) = 4x - 3$.

$$\begin{aligned} f''(x) &= \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(4(x+h) - 3) - (4x - 3)}{h} \\ &= \lim_{h \rightarrow 0} 4 \\ &= 4 \end{aligned}$$

Use $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ with $f'(x)$ in

place of $f(x)$.

Substitute $f'(x+h) = 4(x+h) - 3$ and

$$f'(x) = 4x - 3.$$

Simplify.

Take the limit.

 3.9 Find $f''(x)$ for $f(x) = x^2$.

EXAMPLE 3.16

Finding Acceleration

The position of a particle along a coordinate axis at time t (in seconds) is given by $s(t) = 3t^2 - 4t + 1$ (in meters). Find the function that describes its acceleration at time t .

Solution

Since $v(t) = s'(t)$ and $a(t) = v'(t) = s''(t)$, we begin by finding the derivative of $s(t)$:

$$\begin{aligned}
 s'(t) &= \lim_{h \rightarrow 0} \frac{s(t+h) - s(t)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{3(t+h)^2 - 4(t+h) + 1 - (3t^2 - 4t + 1)}{h} \\
 &= 6t - 4.
 \end{aligned}$$

Next,

$$\begin{aligned}
 s''(t) &= \lim_{h \rightarrow 0} \frac{s'(t+h) - s'(t)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{6(t+h) - 4 - (6t - 4)}{h} \\
 &= 6.
 \end{aligned}$$

Thus, $a = 6 \text{ m/s}^2$.

✓ 3.10 For $s(t) = t^3$, find $a(t)$.



SECTION 3.2 EXERCISES

For the following exercises, use the definition of a derivative to find $f'(x)$.

54. $f(x) = 6$

55. $f(x) = 2 - 3x$

56. $f(x) = \frac{2x}{7} + 1$

57. $f(x) = 4x^2$

58. $f(x) = 5x - x^2$

59. $f(x) = \sqrt{2x}$

60. $f(x) = \sqrt{x - 6}$

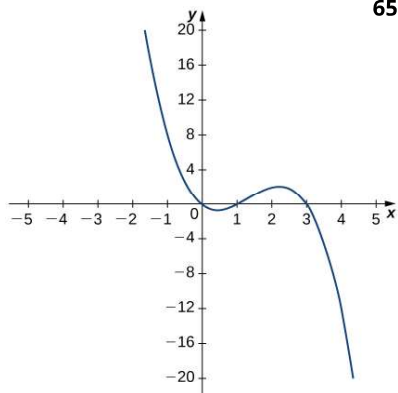
61. $f(x) = \frac{9}{x}$

62. $f(x) = x + \frac{1}{x}$

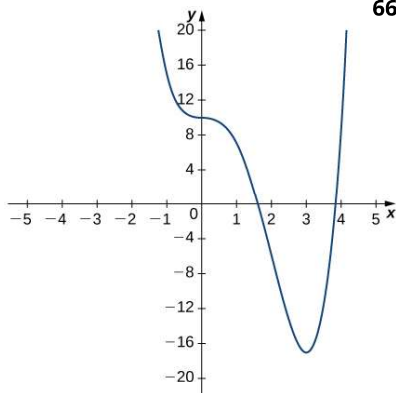
63. $f(x) = \frac{1}{\sqrt{x}}$

For the following exercises, use the graph of $y = f(x)$ to sketch the graph of its derivative $f'(x)$.

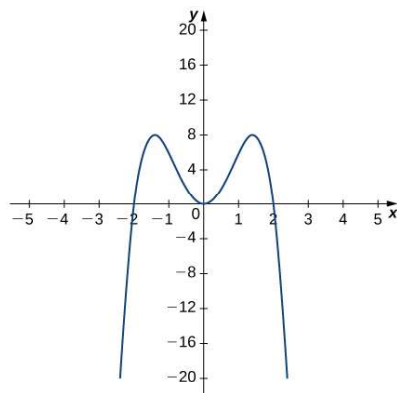
64.



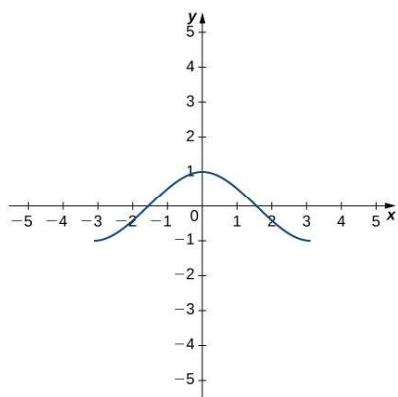
65.



66.



67.



For the following exercises, the given limit represents the derivative of a function $y = f(x)$ at $x = a$. Find $f(x)$ and a .

68. $\lim_{h \rightarrow 0} \frac{(1+h)^{2/3} - 1}{h}$

69. $\lim_{h \rightarrow 0} \frac{[3(2+h)^2 + 2] - 14}{h}$

70. $\lim_{h \rightarrow 0} \frac{\cos(\pi+h) + 1}{h}$

71. $\lim_{h \rightarrow 0} \frac{(2+h)^4 - 16}{h}$

72. $\lim_{h \rightarrow 0} \frac{[2(3+h)^2 - (3+h)] - 15}{h}$

73. $\lim_{h \rightarrow 0} \frac{e^h - 1}{h}$

For the following functions,

- a. sketch the graph and
b. use the definition of a derivative to show that the function is not differentiable at $x = 1$.

74. $f(x) = \begin{cases} 2\sqrt{x}, & 0 \leq x \leq 1 \\ 3x - 1, & x > 1 \end{cases}$

75. $f(x) = \begin{cases} 3, & x < 1 \\ 3x, & x \geq 1 \end{cases}$

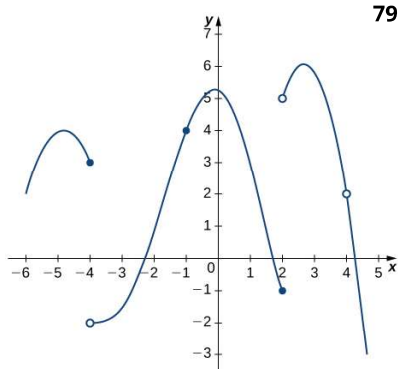
76. $f(x) = \begin{cases} -x^2 + 2, & x \leq 1 \\ x, & x > 1 \end{cases}$

77. $f(x) = \begin{cases} 2x, & x \leq 1 \\ \frac{2}{x}, & x > 1 \end{cases}$

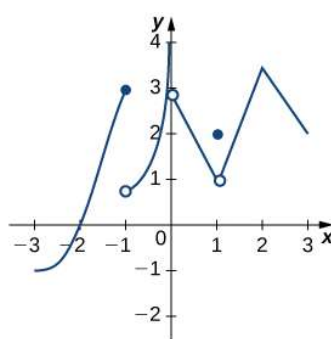
For the following graphs,

- a. determine for which values of $x = a$ the $\lim_{x \rightarrow a} f(x)$ exists but f is not continuous at $x = a$, and
b. determine for which values of $x = a$ the function is continuous but not differentiable at $x = a$.

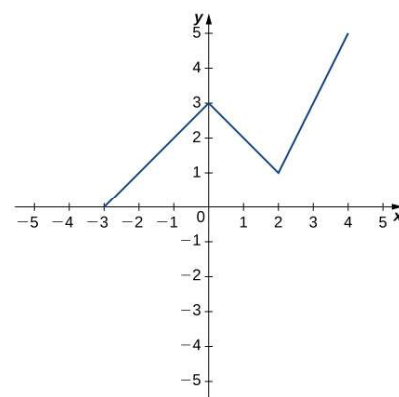
78.



79.



80. Use the graph to evaluate a. $f'(-0.5)$, b. $f'(0)$, c. $f'(1)$, d. $f'(2)$, and e. $f'(3)$, if it exists.



For the following functions, use $f''(x) = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h}$ to find $f''(x)$.

81. $f(x) = 2 - 3x$

82. $f(x) = 4x^2$

83. $f(x) = x + \frac{1}{x}$

For the following exercises, use a calculator to graph $f(x)$. Determine the function $f'(x)$, then use a calculator to graph $f'(x)$.

84. [T] $f(x) = -\frac{5}{x}$

85. [T] $f(x) = 3x^2 + 2x + 4.$

86. [T] $f(x) = \sqrt{x} + 3x$

87. [T] $f(x) = \frac{1}{\sqrt{2x}}$

88. [T] $f(x) = 1 + x + \frac{1}{x}$

89. [T] $f(x) = x^3 + 1$

For the following exercises, describe what the two expressions represent in terms of each of the given situations. Be sure to include units.

a. $\frac{f(x+h)-f(x)}{h}$

b. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$

90. $P(x)$ denotes the population of a city at time x in years.

91. $C(x)$ denotes the total amount of money (in thousands of dollars) spent on concessions by x customers at an amusement park.

92. $R(x)$ denotes the total cost (in thousands of dollars) of manufacturing x clock radios.

93. $g(x)$ denotes the grade (in percentage points) received on a test, given x hours of studying.

94. $B(x)$ denotes the cost (in dollars) of a sociology textbook at university bookstores in the United States in x years since 1990.

95. $p(x)$ denotes atmospheric pressure in Torr at an altitude of x feet.

96. Sketch the graph of a function $y = f(x)$ with all of the following properties:

- $f'(x) > 0$ for $-2 \leq x < 1$
- $f'(2) = 0$
- $f'(x) > 0$ for $x > 2$
- $f(2) = 2$ and $f(0) = 1$
- $\lim_{x \rightarrow -\infty} f(x) = 0$ and $\lim_{x \rightarrow \infty} f(x) = \infty$
- $f'(1)$ does not exist.

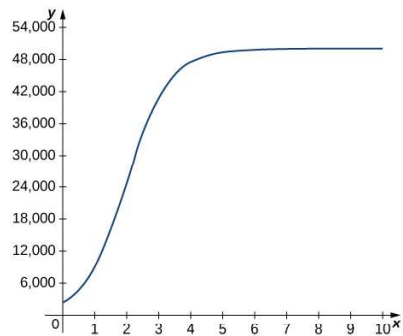
97. Suppose temperature T in degrees Fahrenheit at a height x in feet above the ground is given by $y = T(x)$.

- Give a physical interpretation, with units, of $T'(x)$.
- If we know that $T'(1000) = -0.1$, explain the physical meaning.

98. Suppose the total profit of a company is $y = P(x)$ thousand dollars when x units of an item are sold.

- What does $\frac{P(b)-P(a)}{b-a}$ for $0 < a < b$ measure, and what are the units?
- What does $P'(x)$ measure, and what are the units?
- Suppose that $P'(30) = 5$, what is the approximate change in profit if the number of items sold increases from 30 to 31?

99. The graph in the following figure models the number of people $N(t)$ who have come down with the flu t weeks after its initial outbreak in a town with a population of 50,000 citizens.
- a. Describe what $N'(t)$ represents and how it behaves as t increases.
 - b. What does the derivative tell us about how this town is affected by the flu outbreak?



For the following exercises, use the following table, which shows the height h of the Saturn V rocket for the Apollo 11 mission t seconds after launch.

Time (seconds)	Height (meters)
0	0
1	2
2	4
3	13
4	25
5	32

100. What is the physical meaning of $h'(t)$? What are the units?
101. [T] Construct a table of values for $h'(t)$ and graph both $h(t)$ and $h'(t)$ on the same graph. (*Hint:* for **interior points**, estimate both the left limit and right limit and average them. An interior point of an interval I is an element of I which is not an endpoint of I .)
102. [T] The best linear fit to the data is given by $H(t) = 7.229t - 4.905$, where H is the height of the rocket (in meters) and t is the time elapsed since takeoff. From this equation, determine $H'(t)$. Graph $H(t)$ with the given data and, on a separate coordinate plane, graph $H'(t)$.

- 103. [T]** The best quadratic fit to the data is given by
 $G(t) = 1.429t^2 + 0.0857t - 0.1429$,
 where G is the height of the rocket (in meters) and t is the time elapsed since takeoff. From this equation, determine $G'(t)$. Graph $G(t)$ with the given data and, on a separate coordinate plane, graph $G'(t)$.
- 104. [T]** The best cubic fit to the data is given by
 $F(t) = 0.2037t^3 + 2.956t^2 - 2.705t + 0.4683$,
 where F is the height of the rocket (in m) and t is the time elapsed since take off. From this equation, determine $F'(t)$. Graph $F(t)$ with the given data and, on a separate coordinate plane, graph $F'(t)$. Does the linear, quadratic, or cubic function fit the data best?
- 105.** Using the best linear, quadratic, and cubic fits to the data, determine what $H''(t)$, $G''(t)$ and $F''(t)$ are. What are the physical meanings of $H''(t)$, $G''(t)$ and $F''(t)$, and what are their units?

3.3 Differentiation Rules

Learning Objectives

- 3.3.1** State the constant, constant multiple, and power rules.
3.3.2 Apply the sum and difference rules to combine derivatives.
3.3.3 Use the product rule for finding the derivative of a product of functions.
3.3.4 Use the quotient rule for finding the derivative of a quotient of functions.
3.3.5 Extend the power rule to functions with negative exponents.
3.3.6 Combine the differentiation rules to find the derivative of a polynomial or rational function.

Finding derivatives of functions by using the definition of the derivative can be a lengthy and, for certain functions, a rather challenging process. For example, previously we found that $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$ by using a process that involved

multiplying an expression by a conjugate prior to evaluating a limit. The process that we could use to evaluate $\frac{d}{dx}(\sqrt[3]{x})$ using the definition, while similar, is more complicated. In this section, we develop rules for finding derivatives that allow us to bypass this process. We begin with the basics.

The Basic Rules

The functions $f(x) = c$ and $g(x) = x^n$ where n is a positive integer are the building blocks from which all polynomials and rational functions are constructed. To find derivatives of polynomials and rational functions efficiently without resorting to the limit definition of the derivative, we must first develop formulas for differentiating these basic functions.

The Constant Rule

We first apply the limit definition of the derivative to find the derivative of the constant function, $f(x) = c$. For this function, both $f(x) = c$ and $f(x+h) = c$, so we obtain the following result:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c - c}{h} \\ &= \lim_{h \rightarrow 0} \frac{0}{h} \\ &= \lim_{h \rightarrow 0} 0 = 0. \end{aligned}$$

The rule for differentiating constant functions is called the **constant rule**. It states that the derivative of a constant function is zero; that is, since a constant function is a horizontal line, the slope, or the rate of change, of a constant function is 0. We restate this rule in the following theorem.

Theorem 3.2**The Constant Rule**

Let c be a constant.

If $f(x) = c$, then $f'(x) = 0$.

Alternatively, we may express this rule as

$$\frac{d}{dx}(c) = 0.$$

EXAMPLE 3.17**Applying the Constant Rule**

Find the derivative of $f(x) = 8$.

 Solution

This is just a one-step application of the rule:

$$f'(x) = 0.$$

 3.11 Find the derivative of $g(x) = -3$.

The Power Rule

We have shown that

$$\frac{d}{dx}(x^2) = 2x \text{ and } \frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{-1/2}.$$

At this point, you might see a pattern beginning to develop for derivatives of the form $\frac{d}{dx}(x^n)$. We continue our examination of derivative formulas by differentiating power functions of the form $f(x) = x^n$ where n is a positive integer. We develop formulas for derivatives of this type of function in stages, beginning with positive integer powers. Before stating and proving the general rule for derivatives of functions of this form, we take a look at a specific case, $\frac{d}{dx}(x^3)$. As we go through this derivation, note that the technique used in this case is essentially the same as the technique used to prove the general case.

EXAMPLE 3.18**Differentiating x^3**

Find $\frac{d}{dx}(x^3)$.

 Solution

$$\begin{aligned}
 \frac{d}{dx}(x^3) &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\
 &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} \\
 &= \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h} \\
 &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) \\
 &= 3x^2
 \end{aligned}$$

Notice that the first term in the expansion of $(x + h)^3$ is x^3 and the second term is $3x^2h$. All other terms contain powers of h that are two or greater.

In this step the x^3 terms have been cancelled, leaving only terms containing h .

Factor out the common factor of h .

After cancelling the common factor of h , the only term not containing h is $3x^2$.

Let h go to 0.

 3.12 Find $\frac{d}{dx}(x^4)$.

As we shall see, the procedure for finding the derivative of the general form $f(x) = x^n$ is very similar. Although it is often unwise to draw general conclusions from specific examples, we note that when we differentiate $f(x) = x^3$, the power on x becomes the coefficient of x^2 in the derivative and the power on x in the derivative decreases by 1. The following theorem states that the **power rule** holds for all positive integer powers of x . We will eventually extend this result to negative integer powers. Later, we will see that this rule may also be extended first to rational powers of x and then to arbitrary powers of x . Be aware, however, that this rule does not apply to functions in which a constant is raised to a variable power, such as $f(x) = 3^x$.

Theorem 3.3

The Power Rule

Let n be a positive integer. If $f(x) = x^n$, then

$$f'(x) = nx^{n-1}.$$

Alternatively, we may express this rule as

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Proof

For $f(x) = x^n$ where n is a positive integer, we have

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}.$$

$$\text{Since } (x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + \dots + nxh^{n-1} + h^n,$$

we see that

$$(x+h)^n - x^n = nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + \dots + nxh^{n-1} + h^n.$$

Next, divide both sides by h :

$$\frac{(x+h)^n - x^n}{h} = \frac{nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + \dots + nxh^{n-1} + h^n}{h}.$$

Thus,

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2} x^{n-2}h + \binom{n}{3} x^{n-3}h^2 + \dots + nxh^{n-2} + h^{n-1}.$$

Finally,

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \left(nx^{n-1} + \binom{n}{2} x^{n-2}h + \binom{n}{3} x^{n-3}h^2 + \dots + nxh^{n-2} + h^{n-1} \right) \\ &= nx^{n-1}. \end{aligned}$$

□

EXAMPLE 3.19

Applying the Power Rule

Find the derivative of the function $f(x) = x^{10}$ by applying the power rule.

Solution

Using the power rule with $n = 10$, we obtain

$$f'(x) = 10x^{10-1} = 10x^9.$$

 3.13 Find the derivative of $f(x) = x^7$.

The Sum, Difference, and Constant Multiple Rules

We find our next differentiation rules by looking at derivatives of sums, differences, and constant multiples of functions. Just as when we work with functions, there are rules that make it easier to find derivatives of functions that we add, subtract, or multiply by a constant. These rules are summarized in the following theorem.

Theorem 3.4

Sum, Difference, and Constant Multiple Rules

Let $f(x)$ and $g(x)$ be differentiable functions and k be a constant. Then each of the following equations holds.

Sum Rule. The derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the derivative of g .

$$\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}(f(x)) + \frac{d}{dx}(g(x));$$

that is,

$$\text{for } j(x) = f(x) + g(x), j'(x) = f'(x) + g'(x).$$

Difference Rule. The derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g :

$$\frac{d}{dx}(f(x) - g(x)) = \frac{d}{dx}(f(x)) - \frac{d}{dx}(g(x));$$

that is,

$$\text{for } j(x) = f(x) - g(x), j'(x) = f'(x) - g'(x).$$

Constant Multiple Rule. The derivative of a constant k multiplied by a function f is the same as the constant multiplied by the derivative:

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}(f(x));$$

that is,

$$\text{for } j(x) = kf(x), j'(x) = kf'(x).$$

Proof

We provide only the proof of the sum rule here. The rest follow in a similar manner.

For differentiable functions $f(x)$ and $g(x)$, we set $j(x) = f(x) + g(x)$. Using the limit definition of the derivative we have

$$j'(x) = \lim_{h \rightarrow 0} \frac{j(x+h) - j(x)}{h}.$$

By substituting $j(x+h) = f(x+h) + g(x+h)$ and $j(x) = f(x) + g(x)$, we obtain

$$j'(x) = \lim_{h \rightarrow 0} \frac{(f(x+h) + g(x+h)) - (f(x) + g(x))}{h}.$$

Rearranging and regrouping the terms, we have

$$j'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right).$$

We now apply the sum law for limits and the definition of the derivative to obtain

$$j'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) + \lim_{h \rightarrow 0} \left(\frac{g(x+h) - g(x)}{h} \right) = f'(x) + g'(x).$$

□

EXAMPLE 3.20

Applying the Constant Multiple Rule

Find the derivative of $g(x) = 3x^2$ and compare it to the derivative of $f(x) = x^2$.

Solution

We use the power rule directly:

$$g'(x) = \frac{d}{dx}(3x^2) = 3 \frac{d}{dx}(x^2) = 3(2x) = 6x.$$

Since $f(x) = x^2$ has derivative $f'(x) = 2x$, we see that the derivative of $g(x)$ is 3 times the derivative of $f(x)$. This relationship is illustrated in [Figure 3.18](#).

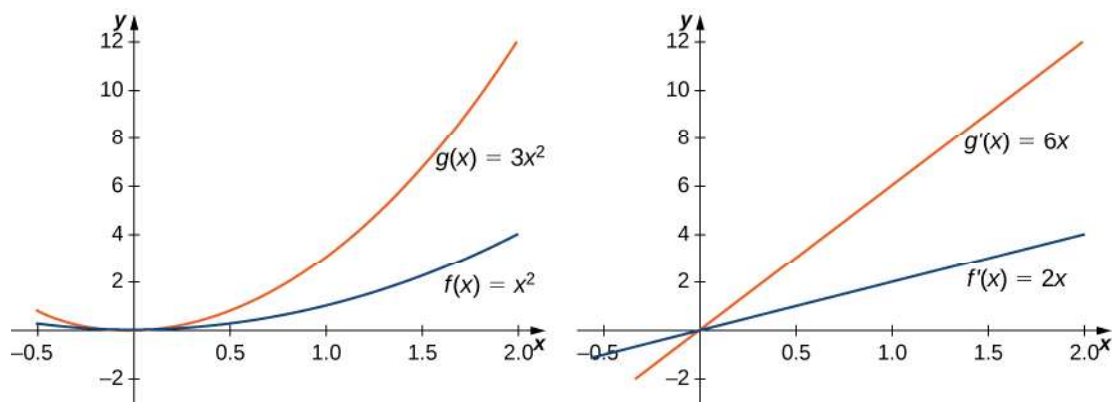


Figure 3.18 The derivative of $g(x)$ is 3 times the derivative of $f(x)$.

EXAMPLE 3.21

Applying Basic Derivative Rules

Find the derivative of $f(x) = 2x^5 + 7$.

✓ **Solution**

We begin by applying the rule for differentiating the sum of two functions, followed by the rules for differentiating constant multiples of functions and the rule for differentiating powers. To better understand the sequence in which the differentiation rules are applied, we use Leibniz notation throughout the solution:

$$\begin{aligned}
 f'(x) &= \frac{d}{dx}(2x^5 + 7) \\
 &= \frac{d}{dx}(2x^5) + \frac{d}{dx}(7) && \text{Apply the sum rule.} \\
 &= 2 \frac{d}{dx}(x^5) + \frac{d}{dx}(7) && \text{Apply the constant multiple rule.} \\
 &= 2(5x^4) + 0 && \text{Apply the power rule and the constant rule.} \\
 &= 10x^4. && \text{Simplify.}
 \end{aligned}$$

✓ 3.14 Find the derivative of $f(x) = 2x^3 - 6x^2 + 3$.

EXAMPLE 3.22

Finding the Equation of a Tangent Line

Find an equation of the line tangent to the graph of $f(x) = x^2 - 4x + 6$ at $x = 1$.

✓ **Solution**

To find an equation of the tangent line, we need a point and a slope. To find the point, compute

$$f(1) = 1^2 - 4(1) + 6 = 3.$$

This gives us the point $(1, 3)$. Since the slope of the tangent line at 1 is $f'(1)$, we must first find $f'(x)$. Using the definition of a derivative, we have

$$f'(x) = 2x - 4$$

so the slope of the tangent line is $f'(1) = -2$. Using the point-slope formula, we see that the equation of the tangent line is

$$y - 3 = -2(x - 1).$$

Putting the equation of the line in slope-intercept form, we obtain

$$y = -2x + 5.$$

✓ 3.15 Find an equation of the line tangent to the graph of $f(x) = 3x^2 - 11$ at $x = 2$. Use the point-slope form.

The Product Rule

Now that we have examined the basic rules, we can begin looking at some of the more advanced rules. The first one examines the derivative of the product of two functions. Although it might be tempting to assume that the derivative of the product is the product of the derivatives, similar to the sum and difference rules, the **product rule** does not follow this pattern. To see why we cannot use this pattern, consider the function $f(x) = x^2$, whose derivative is $f'(x) = 2x$ and not $\frac{d}{dx}(x) \cdot \frac{d}{dx}(x) = 1 \cdot 1 = 1$.

Theorem 3.5

Product Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}(f(x)g(x)) = \frac{d}{dx}(f(x)) \cdot g(x) + \frac{d}{dx}(g(x)) \cdot f(x).$$

That is,

$$\text{if } j(x) = f(x)g(x), \text{ then } j'(x) = f'(x)g(x) + g'(x)f(x).$$

This means that the derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function.

Proof

We begin by assuming that $f(x)$ and $g(x)$ are differentiable functions. At a key point in this proof we need to use the fact that, since $g(x)$ is differentiable, it is also continuous. In particular, we use the fact that since $g(x)$ is continuous,

$$\lim_{h \rightarrow 0} g(x+h) = g(x).$$

By applying the limit definition of the derivative to $j(x) = f(x)g(x)$, we obtain

$$j'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}.$$

By adding and subtracting $f(x)g(x+h)$ in the numerator, we have

$$j'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h}.$$

After breaking apart this quotient and applying the sum law for limits, the derivative becomes

$$j'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h)g(x+h) - f(x)g(x+h)}{h} \right) + \lim_{h \rightarrow 0} \left(\frac{f(x)g(x+h) - f(x)g(x)}{h} \right).$$

Rearranging, we obtain

$$j'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \cdot g(x+h) \right) + \lim_{h \rightarrow 0} \left(\frac{g(x+h) - g(x)}{h} \cdot f(x) \right).$$

By using the continuity of $g(x)$, the definition of the derivatives of $f(x)$ and $g(x)$, and applying the limit laws, we arrive at the product rule,

$$j'(x) = f'(x)g(x) + g'(x)f(x).$$

□

EXAMPLE 3.23

Applying the Product Rule to Functions at a Point

For $j(x) = f(x)g(x)$, use the product rule to find $j'(2)$ if $f(2) = 3$, $f'(2) = -4$, $g(2) = 1$, and $g'(2) = 6$.

Solution

Since $j(x) = f(x)g(x)$, $j'(x) = f'(x)g(x) + g'(x)f(x)$, and hence

$$j'(2) = f'(2)g(2) + g'(2)f(2) = (-4)(1) + (6)(3) = 14.$$

EXAMPLE 3.24

Applying the Product Rule to Binomials

For $j(x) = (x^2 + 2)(3x^3 - 5x)$, find $j'(x)$ by applying the product rule. Check the result by first finding the product and then differentiating.

Solution

If we set $f(x) = x^2 + 2$ and $g(x) = 3x^3 - 5x$, then $f'(x) = 2x$ and $g'(x) = 9x^2 - 5$. Thus,

$$j'(x) = f'(x)g(x) + g'(x)f(x) = (2x)(3x^3 - 5x) + (9x^2 - 5)(x^2 + 2).$$

Simplifying, we have

$$j'(x) = 15x^4 + 3x^2 - 10.$$

To check, we see that $j(x) = 3x^5 + x^3 - 10x$ and, consequently, $j'(x) = 15x^4 + 3x^2 - 10$.

- ✓ 3.16 Use the product rule to obtain the derivative of $j(x) = 2x^5(4x^2 + x)$.

The Quotient Rule

Having developed and practiced the product rule, we now consider differentiating quotients of functions. As we see in the following theorem, the derivative of the quotient is not the quotient of the derivatives; rather, it is the derivative of the function in the numerator times the function in the denominator minus the derivative of the function in the denominator times the function in the numerator, all divided by the square of the function in the denominator. In order to better grasp why we cannot simply take the quotient of the derivatives, keep in mind that

$$\frac{d}{dx}(x^2) = 2x, \text{ not } \frac{\frac{d}{dx}(x^3)}{\frac{d}{dx}(x)} = \frac{3x^2}{1} = 3x^2.$$

Theorem 3.6

The Quotient Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}(f(x)) \cdot g(x) - \frac{d}{dx}(g(x)) \cdot f(x)}{(g(x))^2}.$$

That is,

$$\text{if } j(x) = \frac{f(x)}{g(x)}, \text{ then } j'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2}.$$

The proof of the **quotient rule** is very similar to the proof of the product rule, so it is omitted here. Instead, we apply this new rule for finding derivatives in the next example.

EXAMPLE 3.25

Applying the Quotient Rule

Use the quotient rule to find the derivative of $k(x) = \frac{5x^2}{4x+3}$.

✓ Solution

Let $f(x) = 5x^2$ and $g(x) = 4x + 3$. Thus, $f'(x) = 10x$ and $g'(x) = 4$. Substituting into the quotient rule, we have

$$k'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2} = \frac{10x(4x+3) - 4(5x^2)}{(4x+3)^2}.$$

Simplifying, we obtain

$$k'(x) = \frac{20x^2 + 30x}{(4x+3)^2}.$$

- ✓ 3.17 Find the derivative of $h(x) = \frac{3x+1}{4x-3}$.

It is now possible to use the quotient rule to extend the power rule to find derivatives of functions of the form x^k where k is a negative integer.

Theorem 3.7

Extended Power Rule

If k is a negative integer, then

$$\frac{d}{dx}(x^k) = kx^{k-1}.$$

Proof

If k is a negative integer, we may set $n = -k$, so that n is a positive integer with $k = -n$. Since for each positive integer n , $x^{-n} = \frac{1}{x^n}$, we may now apply the quotient rule by setting $f(x) = 1$ and $g(x) = x^n$. In this case, $f'(x) = 0$ and $g'(x) = nx^{n-1}$. Thus,

$$\frac{d}{dx}(x^{-n}) = \frac{0(x^n) - 1(nx^{n-1})}{(x^n)^2}.$$

Simplifying, we see that

$$\frac{d}{dx}(x^{-n}) = \frac{-nx^{n-1}}{x^{2n}} = -nx^{(n-1)-2n} = -nx^{-n-1}.$$

Finally, observe that since $k = -n$, by substituting we have

$$\frac{d}{dx}(x^k) = kx^{k-1}.$$

□

EXAMPLE 3.26**Using the Extended Power Rule**

Find $\frac{d}{dx}(x^{-4})$.

 Solution

By applying the extended power rule with $k = -4$, we obtain

$$\frac{d}{dx}(x^{-4}) = -4x^{-4-1} = -4x^{-5}.$$

EXAMPLE 3.27**Using the Extended Power Rule and the Constant Multiple Rule**

Use the extended power rule and the constant multiple rule to find the derivative of $f(x) = \frac{6}{x^2}$.

 Solution

It may seem tempting to use the quotient rule to find this derivative, and it would certainly not be incorrect to do so. However, it is far easier to differentiate this function by first rewriting it as $f(x) = 6x^{-2}$.

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(\frac{6}{x^2}\right) = \frac{d}{dx}(6x^{-2}) && \text{Rewrite } \frac{6}{x^2} \text{ as } 6x^{-2}. \\ &= 6\frac{d}{dx}(x^{-2}) && \text{Apply the constant multiple rule.} \\ &= 6(-2x^{-3}) && \text{Use the extended power rule to differentiate } x^{-2}. \\ &= -12x^{-3} && \text{Simplify.} \end{aligned}$$

 3.18 Find the derivative of $g(x) = \frac{1}{x^7}$ using the extended power rule.

Combining Differentiation Rules

As we have seen throughout the examples in this section, it seldom happens that we are called on to apply just one differentiation rule to find the derivative of a given function. At this point, by combining the differentiation rules, we may find the derivatives of any polynomial or rational function. Later on we will encounter more complex combinations of differentiation rules. A good rule of thumb to use when applying several rules is to apply the rules in reverse of the order

in which we would evaluate the function.

EXAMPLE 3.28

Combining Differentiation Rules

For $k(x) = 3h(x) + x^2g(x)$, find $k'(x)$.

Solution

Finding this derivative requires the sum rule, the constant multiple rule, and the product rule.

$$\begin{aligned} k'(x) &= \frac{d}{dx}(3h(x) + x^2g(x)) = \frac{d}{dx}(3h(x)) + \frac{d}{dx}(x^2g(x)) && \text{Apply the sum rule.} \\ &= 3\frac{d}{dx}(h(x)) + \left(\frac{d}{dx}(x^2)g(x) + \frac{d}{dx}(g(x))x^2\right) && \text{Apply the constant multiple rule to} \\ &= 3h'(x) + 2xg(x) + x^2g'(x) && \text{differentiate } 3h(x) \text{ and the product} \\ & && \text{rule to differentiate } x^2g(x). \end{aligned}$$

EXAMPLE 3.29

Extending the Product Rule

For $k(x) = f(x)g(x)h(x)$, express $k'(x)$ in terms of $f(x)$, $g(x)$, $h(x)$, and their derivatives.

Solution

We can think of the function $k(x)$ as the product of the function $f(x)g(x)$ and the function $h(x)$. That is, $k(x) = (f(x)g(x)) \cdot h(x)$. Thus,

$$\begin{aligned} k'(x) &= \frac{d}{dx}(f(x)g(x)) \cdot h(x) + \frac{d}{dx}(h(x)) \cdot (f(x)g(x)) && \text{Apply the product rule to the product} \\ &= (f'(x)g(x) + g'(x)f(x))h(x) + h'(x)f(x)g(x) && \text{of } f(x)g(x) \text{ and } h(x). \\ &= f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x). && \text{Apply the product rule to } f(x)g(x). \\ & && \text{Simplify.} \end{aligned}$$

EXAMPLE 3.30

Combining the Quotient Rule and the Product Rule

For $h(x) = \frac{2x^3k(x)}{3x+2}$, find $h'(x)$.

Solution

This procedure is typical for finding the derivative of a rational function.

$$\begin{aligned} h'(x) &= \frac{\frac{d}{dx}(2x^3k(x)) \cdot (3x+2) - \frac{d}{dx}(3x+2) \cdot (2x^3k(x))}{(3x+2)^2} && \text{Apply the quotient rule.} \\ &= \frac{(6x^2k(x) + k'(x) \cdot 2x^3)(3x+2) - 3(2x^3k(x))}{(3x+2)^2} && \text{Apply the product rule to find} \\ &= \frac{-6x^3k(x) + 18x^3k(x) + 12x^2k(x) + 6x^4k'(x) + 4x^3k'(x)}{(3x+2)^2} && \frac{d}{dx}(2x^3k(x)). \text{ Use } \frac{d}{dx}(3x+2) = 3. \\ &= \frac{12k(x)(x^3+x^2) + 2k'(x)(3x^4+2x^3)}{(3x+2)^2} && \text{Simplify.} \end{aligned}$$

 3.19 Find $\frac{d}{dx}(3f(x) - 2g(x))$.

EXAMPLE 3.31**Determining Where a Function Has a Horizontal Tangent**

Determine the values of x for which $f(x) = x^3 - 7x^2 + 8x + 1$ has a horizontal tangent line.

✓ Solution

To find the values of x for which $f(x)$ has a horizontal tangent line, we must solve $f'(x) = 0$. Since

$$f'(x) = 3x^2 - 14x + 8 = (3x - 2)(x - 4),$$

we must solve $(3x - 2)(x - 4) = 0$. Thus we see that the function has horizontal tangent lines at $x = \frac{2}{3}$ and $x = 4$ as shown in the following graph.

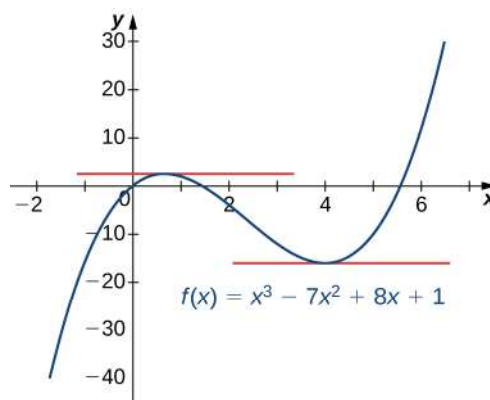


Figure 3.19 This function has horizontal tangent lines at $x = 2/3$ and $x = 4$.

EXAMPLE 3.32**Finding a Velocity**

The position of an object on a coordinate axis at time t is given by $s(t) = \frac{t}{t^2 + 1}$. What is the initial velocity of the object?

✓ Solution

Since the initial velocity is $v(0) = s'(0)$, begin by finding $s'(t)$ by applying the quotient rule:

$$s'(t) = \frac{1(t^2 + 1) - 2t(t)}{(t^2 + 1)^2} = \frac{1 - t^2}{(t^2 + 1)^2}.$$

After evaluating, we see that $v(0) = 1$.

- ✓ 3.20 Find the values of x for which the graph of $f(x) = 4x^2 - 3x + 2$ has a tangent line parallel to the line $y = 2x + 3$.

STUDENT PROJECT**Formula One Grandstands**

Formula One car races can be very exciting to watch and attract a lot of spectators. Formula One track designers have to ensure sufficient grandstand space is available around the track to accommodate these viewers. However, car racing can be dangerous, and safety considerations are paramount. The grandstands must be placed where spectators will not be in danger should a driver lose control of a car ([Figure 3.20](#)).



Figure 3.20 The grandstand next to a straightaway of the Circuit de Barcelona-Catalunya race track, located where the spectators are not in danger.

Safety is especially a concern on turns. If a driver does not slow down enough before entering the turn, the car may slide off the racetrack. Normally, this just results in a wider turn, which slows the driver down. But if the driver loses control completely, the car may fly off the track entirely, on a path tangent to the curve of the racetrack.

Suppose you are designing a new Formula One track. One section of the track can be modeled by the function $f(x) = x^3 + 3x^2 + x$ (Figure 3.21). The current plan calls for grandstands to be built along the first straightaway and around a portion of the first curve. The plans call for the front corner of the grandstand to be located at the point $(-1.9, 2.8)$. We want to determine whether this location puts the spectators in danger if a driver loses control of the car.

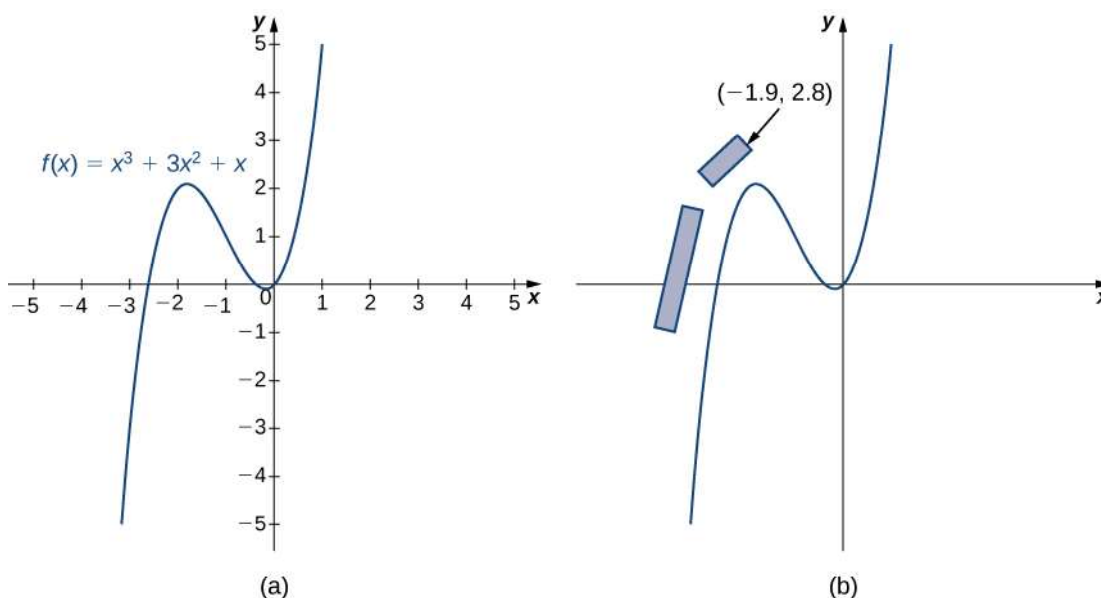


Figure 3.21 (a) One section of the racetrack can be modeled by the function $f(x) = x^3 + 3x^2 + x$. (b) The front corner of the grandstand is located at $(-1.9, 2.8)$.

1. Physicists have determined that drivers are most likely to lose control of their cars as they are coming into a turn, at the point where the slope of the tangent line is 1. Find the (x, y) coordinates of this point near the turn.
2. Find an equation of the tangent line to the curve at this point.

3. To determine whether the spectators are in danger in this scenario, find the x -coordinate of the point where the tangent line crosses the line $y = 2.8$. Is this point safely to the right of the grandstand? Or are the spectators in danger?
4. What if a driver loses control earlier than the physicists project? Suppose a driver loses control at the point $(-2.5, 0.625)$. What is the slope of the tangent line at this point?
5. If a driver loses control as described in part 4, are the spectators safe?
6. Should you proceed with the current design for the grandstand, or should the grandstands be moved?



SECTION 3.3 EXERCISES

For the following exercises, find $f'(x)$ for each function.

106. $f(x) = x^7 + 10$ 107. $f(x) = 5x^3 - x + 1$ 108. $f(x) = 4x^2 - 7x$
 109. $f(x) = 8x^4 + 9x^2 - 1$ 110. $f(x) = x^4 + \frac{2}{x}$ 111. $f(x) = 3x \left(18x^4 + \frac{13}{x+1} \right)$
 112. $f(x) = (x+2)(2x^2 - 3)$ 113. $f(x) = x^2 \left(\frac{2}{x^2} + \frac{5}{x^3} \right)$ 114. $f(x) = \frac{x^3 + 2x^2 - 4}{3}$
 115. $f(x) = \frac{4x^3 - 2x + 1}{x^2}$ 116. $f(x) = \frac{x^2 + 4}{x^2 - 4}$ 117. $f(x) = \frac{x + 9}{x^2 - 7x + 1}$

For the following exercises, find an equation of the tangent line $T(x)$ to the graph of the given function at the indicated point. Use a graphing calculator to graph the function and the tangent line.

118. [T] $y = 3x^2 + 4x + 1$ at $(0, 1)$ 119. [T] $y = \frac{2}{x^2} + 1$ at $(1, 3)$ 120. [T] $y = \frac{2x}{x-1}$ at $(-1, 1)$
 121. [T] $y = \frac{2}{x} - \frac{3}{x^2}$ at $(1, -1)$

For the following exercises, assume that $f(x)$ and $g(x)$ are both differentiable functions for all x . Find the derivative of each of the functions $h(x)$.

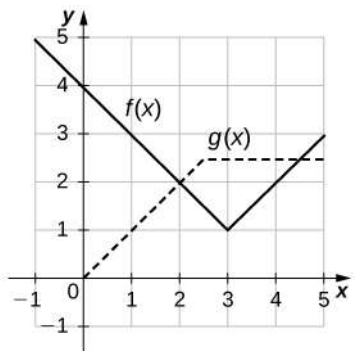
122. $h(x) = 4f(x) + \frac{g(x)}{7}$ 123. $h(x) = x^3 f(x)$ 124. $h(x) = \frac{f(x)g(x)}{2}$
 125. $h(x) = \frac{3f(x)}{g(x)+2}$

For the following exercises, assume that $f(x)$ and $g(x)$ are both differentiable functions with values as given in the following table. Use the following table to calculate the following derivatives.

x	1	2	3	4
$f(x)$	3	5	-2	0
$g(x)$	2	3	-4	6
$f'(x)$	-1	7	8	-3
$g'(x)$	4	1	2	9

126. Find $h'(1)$ if $h(x) = xf(x) + 4g(x)$.
 127. Find $h'(2)$ if $h(x) = \frac{f(x)}{g(x)}$.
 128. Find $h'(3)$ if $h(x) = 2x + f(x)g(x)$.
 129. Find $h'(4)$ if $h(x) = \frac{1}{x} + \frac{g(x)}{f(x)}$.

For the following exercises, use the following figure to find the indicated derivatives, if they exist.



130. Let $h(x) = f(x) + g(x)$. Find
- $h'(1)$,
 - $h'(3)$, and
 - $h'(4)$.
131. Let $h(x) = f(x)g(x)$. Find
- $h'(1)$,
 - $h'(3)$, and
 - $h'(4)$.
132. Let $h(x) = \frac{f(x)}{g(x)}$. Find
- $h'(1)$,
 - $h'(3)$, and
 - $h'(4)$.

For the following exercises,

- evaluate $f'(a)$, and
 - graph the function $f(x)$ and the tangent line at $x = a$.
133. [T] $f(x) = 2x^3 + 3x - x^2$, $a = 2$
134. [T] $f(x) = \frac{1}{x} - x^2$, $a = 1$
135. [T] $f(x) = x^2 - x^{12} + 3x + 2$, $a = 0$
136. [T] $f(x) = \frac{1}{x} - x^2$, $a = -1$
137. Find an equation of the tangent line to the graph of $f(x) = 2x^3 + 4x^2 - 5x - 3$ at $x = -1$.
138. Find an equation of the tangent line to the graph of $f(x) = x^2 + \frac{4}{x} - 10$ at $x = 8$.
139. Find an equation of the tangent line to the graph of $f(x) = (3x - x^2)(3 - x - x^2)$ at $x = 1$.
140. Find the point on the graph of $f(x) = x^3$ such that the tangent line at that point has an x -intercept of 6.
141. Find an equation of the line passing through the point $P(3, 3)$ and tangent to the graph of $f(x) = \frac{6}{x-1}$.
142. Determine all points on the graph of $f(x) = x^3 + x^2 - x - 1$ for which
- the tangent line is horizontal
 - the tangent line has a slope of -1 .
143. Find a quadratic polynomial such that $f(1) = 5$, $f'(1) = 3$ and $f''(1) = -6$.
144. A car driving along a freeway with traffic has traveled $s(t) = t^3 - 6t^2 + 9t$ meters in t seconds.
- Determine the time in seconds when the velocity of the car is 0.
 - Determine the acceleration of the car when the velocity is 0.

- 145. [T]** A herring swimming along a straight line has traveled $s(t) = \frac{t^2}{t^2+2}$ feet in t seconds. Determine the velocity of the herring when it has traveled 3 seconds.
- 146.** The population in millions of arctic flounder in the Atlantic Ocean is modeled by the function $P(t) = \frac{8t+3}{0.2t^2+1}$, where t is measured in years.
- Determine the initial flounder population.
 - Determine $P'(10)$ and briefly interpret the result.
- 147. [T]** The concentration of antibiotic in the bloodstream t hours after being injected is given by the function $C(t) = \frac{2t^2+t}{t^3+50}$, where C is measured in milligrams per liter of blood.
- Find the rate of change of $C(t)$.
 - Determine the rate of change for $t = 8, 12, 24$, and 36 .
 - Briefly describe what seems to be occurring as the number of hours increases.
- 148.** A book publisher has a cost function given by $C(x) = \frac{x^3+2x+3}{x^2}$, where x is the number of copies of a book in thousands and C is the cost, per book, measured in dollars. Evaluate $C'(2)$ and explain its meaning.
- 149. [T]** According to Newton's law of universal gravitation, the force F between two bodies of constant mass m_1 and m_2 is given by the formula $F = \frac{Gm_1m_2}{d^2}$, where G is the gravitational constant and d is the distance between the bodies.
- Suppose that G, m_1 , and m_2 are constants. Find the rate of change of force F with respect to distance d .
 - Find the rate of change of force F with gravitational constant $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, on two bodies 10 meters apart, each with a mass of 1000 kilograms.

3.4 Derivatives as Rates of Change

Learning Objectives

- 3.4.1** Determine a new value of a quantity from the old value and the amount of change.
- 3.4.2** Calculate the average rate of change and explain how it differs from the instantaneous rate of change.
- 3.4.3** Apply rates of change to displacement, velocity, and acceleration of an object moving along a straight line.
- 3.4.4** Predict the future population from the present value and the population growth rate.
- 3.4.5** Use derivatives to calculate marginal cost and revenue in a business situation.

In this section we look at some applications of the derivative by focusing on the interpretation of the derivative as the rate of change of a function. These applications include **acceleration** and velocity in physics, **population growth rates** in biology, and marginal functions in economics.

Amount of Change Formula

One application for derivatives is to estimate an unknown value of a function at a point by using a known value of a function at some given point together with its rate of change at the given point. If $f(x)$ is a function defined on an interval $[a, a + h]$, then the **amount of change** of $f(x)$ over the interval is the change in the y values of the function over that interval and is given by

$$f(a + h) - f(a).$$

The **average rate of change** of the function f over that same interval is the ratio of the amount of change over that interval to the corresponding change in the x values. It is given by

$$\frac{f(a + h) - f(a)}{h}.$$

As we already know, the instantaneous rate of change of $f(x)$ at a is its derivative

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}.$$

For small enough values of h , $f'(a) \approx \frac{f(a+h)-f(a)}{h}$. We can then solve for $f(a + h)$ to get the amount of change formula:

$$f(a + h) \approx f(a) + f'(a)h. \quad (3.10)$$

We can use this formula if we know only $f(a)$ and $f'(a)$ and wish to estimate the value of $f(a + h)$. For example, we may use the current population of a city and the rate at which it is growing to estimate its population in the near future. As we can see in Figure 3.22, we are approximating $f(a + h)$ by the y coordinate at $a + h$ on the line tangent to $f(x)$ at $x = a$. Observe that the accuracy of this estimate depends on the value of h as well as the value of $f'(a)$.

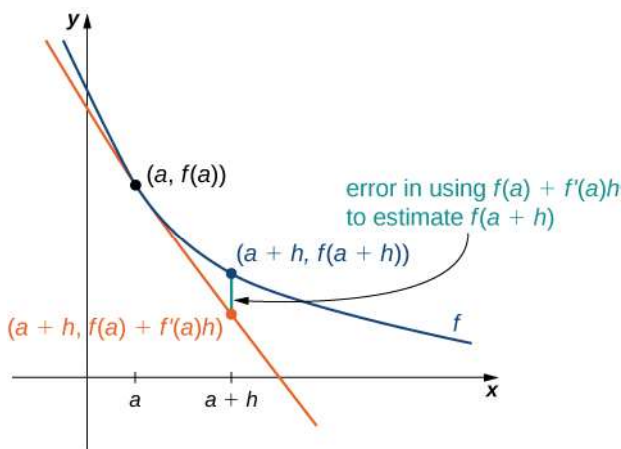


Figure 3.22 The new value of a changed quantity equals the original value plus the rate of change times the interval of change: $f(a + h) \approx f(a) + f'(a)h$.

MEDIA

Here is an interesting [demonstration \(http://www.openstax.org/l/20_chainrule\)](http://www.openstax.org/l/20_chainrule) of rate of change.

EXAMPLE 3.33

Estimating the Value of a Function

If $f(3) = 2$ and $f'(3) = 5$, estimate $f(3.2)$.

Solution

Begin by finding h . We have $h = 3.2 - 3 = 0.2$. Thus,

$$f(3.2) = f(3 + 0.2) \approx f(3) + (0.2)f'(3) = 2 + 0.2(5) = 3.$$

 3.21 Given $f(10) = -5$ and $f'(10) = 6$, estimate $f(10.1)$.

Motion along a Line

Another use for the derivative is to analyze motion along a line. We have described velocity as the rate of change of position. If we take the derivative of the velocity, we can find the acceleration, or the rate of change of velocity. It is also important to introduce the idea of **speed**, which is the magnitude of velocity. Thus, we can state the following mathematical definitions.

Definition

Let $s(t)$ be a function giving the position of an object at time t .

The velocity of the object at time t is given by $v(t) = s'(t)$.

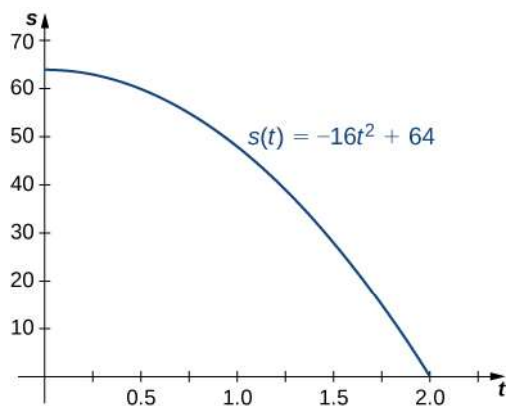
The speed of the object at time t is given by $|v(t)|$.

The acceleration of the object at t is given by $a(t) = v'(t) = s''(t)$.

EXAMPLE 3.34

Comparing Instantaneous Velocity and Average Velocity

A ball is dropped from a height of 64 feet. Its height above ground (in feet) t seconds later is given by $s(t) = -16t^2 + 64$.



- What is the instantaneous velocity of the ball when it hits the ground?
- What is the average velocity during its fall?

Solution

The first thing to do is determine how long it takes the ball to reach the ground. To do this, set $s(t) = 0$. Solving $-16t^2 + 64 = 0$, we get $t = 2$, so it takes 2 seconds for the ball to reach the ground.

- The instantaneous velocity of the ball as it strikes the ground is $v(2)$. Since $v(t) = s'(t) = -32t$, we obtain $v(2) = -64$ ft/s.
- The average velocity of the ball during its fall is

$$v_{ave} = \frac{s(2) - s(0)}{2 - 0} = \frac{0 - 64}{2} = -32 \text{ ft/s.}$$

EXAMPLE 3.35

Interpreting the Relationship between $v(t)$ and $a(t)$

A particle moves along a coordinate axis in the positive direction to the right. Its position at time t is given by $s(t) = t^3 - 4t + 2$. Find $v(1)$ and $a(1)$ and use these values to answer the following questions.

- Is the particle moving from left to right or from right to left at time $t = 1$?
- Is the particle speeding up or slowing down at time $t = 1$?

 Solution

Begin by finding $v(t)$ and $a(t)$.

$$v(t) = s'(t) = 3t^2 - 4 \text{ and } a(t) = v'(t) = s''(t) = 6t.$$

Evaluating these functions at $t = 1$, we obtain $v(1) = -1$ and $a(1) = 6$.

- Because $v(1) < 0$, the particle is moving from right to left.
- Because $v(1) < 0$ and $a(1) > 0$, velocity and acceleration are acting in opposite directions. In other words, the particle is being accelerated in the direction opposite the direction in which it is traveling, causing $|v(t)|$ to decrease. The particle is slowing down.

EXAMPLE 3.36

Position and Velocity

The position of a particle moving along a coordinate axis is given by $s(t) = t^3 - 9t^2 + 24t + 4$, $t \geq 0$.

- Find $v(t)$.
- At what time(s) is the particle at rest?
- On what time intervals is the particle moving from left to right? From right to left?
- Use the information obtained to sketch the path of the particle along a coordinate axis.

 Solution

- The velocity is the derivative of the position function:

$$v(t) = s'(t) = 3t^2 - 18t + 24.$$

- The particle is at rest when $v(t) = 0$, so set $3t^2 - 18t + 24 = 0$. Factoring the left-hand side of the equation produces $3(t - 2)(t - 4) = 0$. Solving, we find that the particle is at rest at $t = 2$ and $t = 4$.
- The particle is moving from left to right when $v(t) > 0$ and from right to left when $v(t) < 0$. [Figure 3.23](#) gives the analysis of the sign of $v(t)$ for $t \geq 0$, but it does not represent the axis along which the particle is moving.



Figure 3.23 The sign of $v(t)$ determines the direction of the particle.

Since $3t^2 - 18t + 24 > 0$ on $[0, 2) \cup (4, +\infty)$, the particle is moving from left to right on these intervals.

Since $3t^2 - 18t + 24 < 0$ on $(2, 4)$, the particle is moving from right to left on this interval.

- Before we can sketch the graph of the particle, we need to know its position at the time it starts moving ($t = 0$) and at the times that it changes direction ($t = 2, 4$). We have $s(0) = 4$, $s(2) = 24$, and $s(4) = 20$. This means that the particle begins on the coordinate axis at 4 and changes direction at 24 and 20 on the coordinate axis. The path of the particle is shown on a coordinate axis in [Figure 3.24](#).

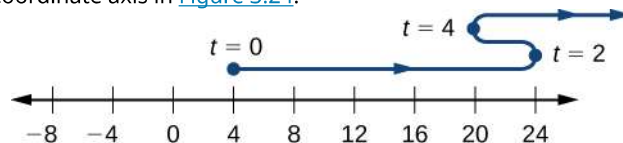


Figure 3.24 The path of the particle can be determined by analyzing $v(t)$.

-  3.22 A particle moves along a coordinate axis. Its position at time t is given by $s(t) = t^2 - 5t + 1$. Is the particle moving from right to left or from left to right at time $t = 3$?

Population Change

In addition to analyzing velocity, speed, acceleration, and position, we can use derivatives to analyze various types of populations, including those as diverse as bacteria colonies and cities. We can use a current population, together with a growth rate, to estimate the size of a population in the future. The population growth rate is the rate of change of a population and consequently can be represented by the derivative of the size of the population.

Definition

If $P(t)$ is the number of entities present in a population, then the population growth rate of $P(t)$ is defined to be $P'(t)$.

EXAMPLE 3.37**Estimating a Population**

The population of a city is tripling every 5 years. If its current population is 10,000, what will be its approximate population 2 years from now?

 Solution

Let $P(t)$ be the population (in thousands) t years from now. Thus, we know that $P(0) = 10$ and based on the information, we anticipate $P(5) = 30$. Now estimate $P'(0)$, the current growth rate, using

$$P'(0) \approx \frac{P(5) - P(0)}{5 - 0} = \frac{30 - 10}{5} = 4.$$

By applying [Equation 3.10](#) to $P(t)$, we can estimate the population 2 years from now by writing

$$P(2) \approx P(0) + (2) P'(0) \approx 10 + 2(4) = 18;$$

thus, in 2 years the population will be 18,000.

-  3.23 The current population of a mosquito colony is known to be 3,000; that is, $P(0) = 3,000$. If $P'(0) = 100$, estimate the size of the population in 3 days, where t is measured in days.

Changes in Cost and Revenue

In addition to analyzing motion along a line and population growth, derivatives are useful in analyzing changes in cost, revenue, and profit. The concept of a marginal function is common in the fields of business and economics and implies the use of derivatives. The marginal cost is the derivative of the cost function. The marginal revenue is the derivative of the revenue function. The marginal profit is the derivative of the profit function, which is based on the cost function and the revenue function.

Definition

If $C(x)$ is the cost of producing x items, then the **marginal cost** $MC(x)$ is $MC(x) = C'(x)$.

If $R(x)$ is the revenue obtained from selling x items, then the **marginal revenue** $MR(x)$ is $MR(x) = R'(x)$.

If $P(x) = R(x) - C(x)$ is the profit obtained from selling x items, then the **marginal profit** $MP(x)$ is defined to be $MP(x) = P'(x) = MR(x) - MC(x) = R'(x) - C'(x)$.

We can roughly approximate

$$MC(x) = C'(x) = \lim_{h \rightarrow 0} \frac{C(x+h) - C(x)}{h}$$

by choosing an appropriate value for h . Since x represents objects, a reasonable and small value for h is 1. Thus, by substituting $h = 1$, we get the approximation $MC(x) = C'(x) \approx C(x+1) - C(x)$. Consequently, $C'(x)$ for a given value of x can be thought of as the change in cost associated with producing one additional item. In a similar way, $MR(x) = R'(x)$ approximates the revenue obtained by selling one additional item, and $MP(x) = P'(x)$ approximates the profit obtained by producing and selling one additional item.

EXAMPLE 3.38**Applying Marginal Revenue**

Assume that the number of barbeque dinners that can be sold, x , can be related to the price charged, p , by the equation $p(x) = 9 - 0.03x$, $0 \leq x \leq 300$.

In this case, the revenue in dollars obtained by selling x barbeque dinners is given by

$$R(x) = xp(x) = x(9 - 0.03x) = -0.03x^2 + 9x \text{ for } 0 \leq x \leq 300.$$

Use the marginal revenue function to estimate the revenue obtained from selling the 101st barbeque dinner. Compare this to the actual revenue obtained from the sale of this dinner.

 Solution

First, find the marginal revenue function: $MR(x) = R'(x) = -0.06x + 9$.

Next, use $R'(100)$ to approximate $R(101) - R(100)$, the revenue obtained from the sale of the 101st dinner. Since $R'(100) = 3$, the revenue obtained from the sale of the 101st dinner is approximately \$3.

The actual revenue obtained from the sale of the 101st dinner is

$$R(101) - R(100) = 602.97 - 600 = 2.97, \text{ or } \$2.97.$$

The marginal revenue is a fairly good estimate in this case and has the advantage of being easy to compute.



- 3.24 Suppose that the profit obtained from the sale of x fish-fry dinners is given by $P(x) = -0.03x^2 + 8x - 50$. Use the marginal profit function to estimate the profit from the sale of the 101st fish-fry dinner.

**SECTION 3.4 EXERCISES**

For the following exercises, the given functions represent the position of a particle traveling along a horizontal line; $t \geq 0$.

- Find the velocity and acceleration functions.
- Determine the time intervals when the object is slowing down or speeding up.

150. $s(t) = 2t^3 - 3t^2 - 12t + 8$ 151. $s(t) = 2t^3 - 15t^2 + 36t - 10$ 152. $s(t) = \frac{t}{1+t^2}$

153. A model rocket is fired vertically upward from the ground. The distance s in feet that the rocket travels from the ground after t seconds is given by $s(t) = -16t^2 + 560t$.
- Find the velocity of the rocket 3 seconds after being fired.
 - Find the acceleration of the rocket 3 seconds after being fired.

154. A ball is thrown downward with a speed of 8 ft/s from the top of a 64-foot-tall building. After t seconds, its height above the ground is given by $s(t) = -16t^2 - 8t + 64$.
- Determine how long it takes for the ball to hit the ground.
 - Determine the velocity of the ball when it hits the ground.

155. The position function $s(t) = t^2 - 3t - 4$ represents the position of the back of a car backing out of a driveway and then driving in a straight line, where s is in feet and t is in seconds. In this case, $s(t) = 0$ represents the time at which the back of the car is at the garage door, so $s(0) = -4$ is the starting position of the car, 4 feet inside the garage.
- Determine the velocity of the car when $s(t) = 0$.
 - Determine the velocity of the car when $s(t) = 14$.

156. The position of a hummingbird flying along a straight line in t seconds is given by $s(t) = 3t^3 - 7t$ meters.

- Determine the velocity of the bird at $t = 1$ sec.
- Determine the acceleration of the bird at $t = 1$ sec.
- Determine the acceleration of the bird when the velocity equals 0.

157. A potato is launched vertically upward with an initial velocity of 100 ft/s from a potato gun at the top of an 85-foot-tall building. The position of the potato relative to the ground after t seconds is given by

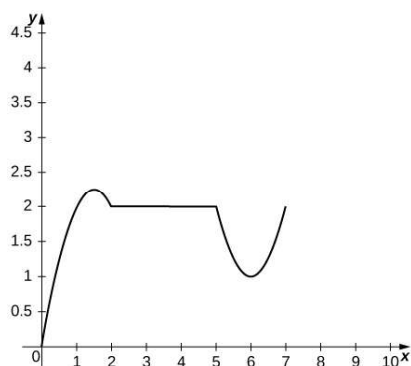
$$s(t) = -16t^2 + 100t + 85.$$

- Find the velocity of the potato after 0.5 s and 5.75 s.
- Find the speed of the potato at 0.5 s and 5.75 s.
- Determine when the potato reaches its maximum height.
- Find the acceleration of the potato at 0.5 s and 1.5 s.
- Determine how long the potato is in the air.
- Determine the velocity of the potato upon hitting the ground.

158. The position function $s(t) = t^3 - 8t$ gives the position in miles of a freight train where east is the positive direction and t is measured in hours.

- Determine the direction the train is traveling when $s(t) = 0$.
- Determine the direction the train is traveling when $a(t) = 0$.
- Determine the time intervals when the train is slowing down or speeding up.

159. The following graph shows the position $y = s(t)$ of an object moving along a straight line.



- Use the graph of the position function to determine the time intervals when the velocity is positive, negative, or zero.
- Sketch the graph of the velocity function.
- Use the graph of the velocity function to determine the time intervals when the acceleration is positive, negative, or zero.
- Determine the time intervals when the object is speeding up or slowing down.

160. The cost function, in dollars, of a company that manufactures food processors is given by $C(x) = 200 + \frac{7}{x} + \frac{x^2}{7}$, where x is the number of food processors manufactured.

- Find the marginal cost function.
- Use the marginal cost function to estimate the cost of manufacturing the thirteenth food processor.
- Find the actual cost of manufacturing the thirteenth food processor.

161. The price p (in dollars) and the demand x for a certain digital clock radio is given by the price-demand function $p = 10 - 0.001x$.

- Find the revenue function $R(x)$.
- Find the marginal revenue function.
- Find the marginal revenue at $x = 2000$ and 5000.

- 162. [T]** A profit is earned when revenue exceeds cost. Suppose the profit function for a skateboard manufacturer is given by $P(x) = 30x - 0.3x^2 - 250$, where x is the number of skateboards sold.
- Find the exact profit from the sale of the thirtieth skateboard.
 - Find the marginal profit function and use it to estimate the profit from the sale of the thirtieth skateboard.
- 163. [T]** In general, the profit function is the difference between the revenue and cost functions:
 $P(x) = R(x) - C(x)$.
 Suppose the price-demand and cost functions for the production of cordless drills is given respectively by $p = 143 - 0.03x$ and $C(x) = 75,000 + 65x$, where x is the number of cordless drills that are sold at a price of p dollars per drill and $C(x)$ is the cost of producing x cordless drills.
- Find the marginal cost function.
 - Find the revenue and marginal revenue functions.
 - Find $R'(1000)$ and $R'(4000)$. Interpret the results.
 - Find the profit and marginal profit functions.
 - Find $P'(1000)$ and $P'(4000)$. Interpret the results.
- 164.** A small town in Ohio commissioned an actuarial firm to conduct a study that modeled the rate of change of the town's population. The study found that the town's population (measured in thousands of people) can be modeled by the function $P(t) = -\frac{1}{3}t^3 + 64t + 3000$, where t is measured in years.
- Find the rate of change function $P'(t)$ of the population function.
 - Find $P'(1)$, $P'(2)$, $P'(3)$, and $P'(4)$. Interpret what the results mean for the town.
 - Find $P''(1)$, $P''(2)$, $P''(3)$, and $P''(4)$. Interpret what the results mean for the town's population.
- 165. [T]** A culture of bacteria grows in number according to the function
 $N(t) = 3000 \left(1 + \frac{4t}{t^2 + 100} \right)$,
 where t is measured in hours.
- Find the rate of change of the number of bacteria.
 - Find $N'(0)$, $N'(10)$, $N'(20)$, and $N'(30)$.
 - Interpret the results in (b).
 - Find $N''(0)$, $N''(10)$, $N''(20)$, and $N''(30)$. Interpret what the answers imply about the bacteria population growth.
- 166.** The centripetal force of an object of mass m is given by $F(r) = \frac{mv^2}{r}$, where v is the speed of rotation and r is the distance from the center of rotation.
- Find the rate of change of centripetal force with respect to the distance from the center of rotation.
 - Find the rate of change of centripetal force of an object with mass 1000 kilograms, velocity of 13.89 m/s, and a distance from the center of rotation of 200 meters.

The following questions concern the population (in millions) of London by decade in the 19th century, which is listed in the following table.

Years since 1800	Population (millions)
1	0.8795
11	1.040
21	1.264
31	1.516
41	1.661
51	2.000
61	2.634
71	3.272
81	3.911
91	4.422

Table 3.4 Population of London Source:

http://en.wikipedia.org/wiki/Demographics_of_London.

167. [T]

- Using a calculator or a computer program, find the best-fit linear function to measure the population.
- Find the derivative of the equation in a. and explain its physical meaning.
- Find the second derivative of the equation and explain its physical meaning.

168. [T]

- Using a calculator or a computer program, find the best-fit quadratic curve through the data.
- Find the derivative of the equation and explain its physical meaning.
- Find the second derivative of the equation and explain its physical meaning.

For the following exercises, consider an astronaut on a large planet in another galaxy. To learn more about the composition of this planet, the astronaut drops an electronic sensor into a deep trench. The sensor transmits its vertical position every second in relation to the astronaut's position. The summary of the falling sensor data is displayed in the following table.

Time after dropping (s)	Position (m)
0	0
1	-1

Time after dropping (s)	Position (m)
2	-2
3	-5
4	-7
5	-14

169. [T]

- Using a calculator or computer program, find the best-fit quadratic curve to the data.
- Find the derivative of the position function and explain its physical meaning.
- Find the second derivative of the position function and explain its physical meaning.

170. [T]

- Using a calculator or computer program, find the best-fit cubic curve to the data.
- Find the derivative of the position function and explain its physical meaning.
- Find the second derivative of the position function and explain its physical meaning.
- Using the result from c. explain why a cubic function is not a good choice for this problem.

The following problems deal with the Holling type I, II, and III equations. These equations describe the ecological event of growth of a predator population given the amount of prey available for consumption.

- 171. [T]** The Holling type I equation is described by $f(x) = ax$, where x is the amount of prey available and $a > 0$ is the rate at which the predator meets the prey for consumption.
- Graph the Holling type I equation, given $a = 0.5$.
 - Determine the first derivative of the Holling type I equation and explain physically what the derivative implies.
 - Determine the second derivative of the Holling type I equation and explain physically what the derivative implies.
 - Using the interpretations from b. and c. explain why the Holling type I equation may not be realistic.
- 172. [T]** The Holling type II equation is described by $f(x) = \frac{ax}{n+x}$, where x is the amount of prey available and $a > 0$ is the maximum consumption rate of the predator.
- Graph the Holling type II equation given $a = 0.5$ and $n = 5$. What are the differences between the Holling type I and II equations?
 - Take the first derivative of the Holling type II equation and interpret the physical meaning of the derivative.
 - Show that $f'(n) = \frac{1}{2}a$ and interpret the meaning of the parameter n .
 - Find and interpret the meaning of the second derivative. What makes the Holling type II function more realistic than the Holling type I function?
- 173. [T]** The Holling type III equation is described by $f(x) = \frac{ax^2}{n^2+x^2}$, where x is the amount of prey available and $a > 0$ is the maximum consumption rate of the predator.
- Graph the Holling type III equation given $a = 0.5$ and $n = 5$. What are the differences between the Holling type II and III equations?
 - Take the first derivative of the Holling type III equation and interpret the physical meaning of the derivative.
 - Find and interpret the meaning of the second derivative (it may help to graph the second derivative).
 - What additional ecological phenomena does the Holling type III function describe compared with the Holling type II function?

174. [T] The populations of the snowshoe hare (in thousands) and the lynx (in hundreds) collected over 7 years from 1937 to 1943 are shown in the following table. The snowshoe hare is the primary prey of the lynx.

Population of snowshoe hare (thousands)	Population of lynx (hundreds)
20	10
55	15
65	55
95	60

Table 3.5 Snowshoe Hare and Lynx Populations

Source:

<http://www.biotopics.co.uk/newgcse/predatorprey.html>.

- Graph the data points and determine which Holling-type function fits the data best.
- Using the meanings of the parameters a and n , determine values for those parameters by examining a graph of the data. Recall that n measures what prey value results in the half-maximum of the predator value.
- Plot the resulting Holling-type I, II, and III functions on top of the data. Was the result from part a. correct?

3.5 Derivatives of Trigonometric Functions

Learning Objectives

- 3.5.1** Find the derivatives of the sine and cosine function.
3.5.2 Find the derivatives of the standard trigonometric functions.
3.5.3 Calculate the higher-order derivatives of the sine and cosine.

One of the most important types of motion in physics is simple harmonic motion, which is associated with such systems as an object with mass oscillating on a spring. Simple harmonic motion can be described by using either sine or cosine